

Autonomous Underwater Vehicle

Grippers for Underwater Manipulation

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Our Goal

Develop an Autonomous Underwater Vehicle (AUV) that has the following capabilities:

01 Autonomous Navigation

- Navigate from Point A to B
- Depth Holding

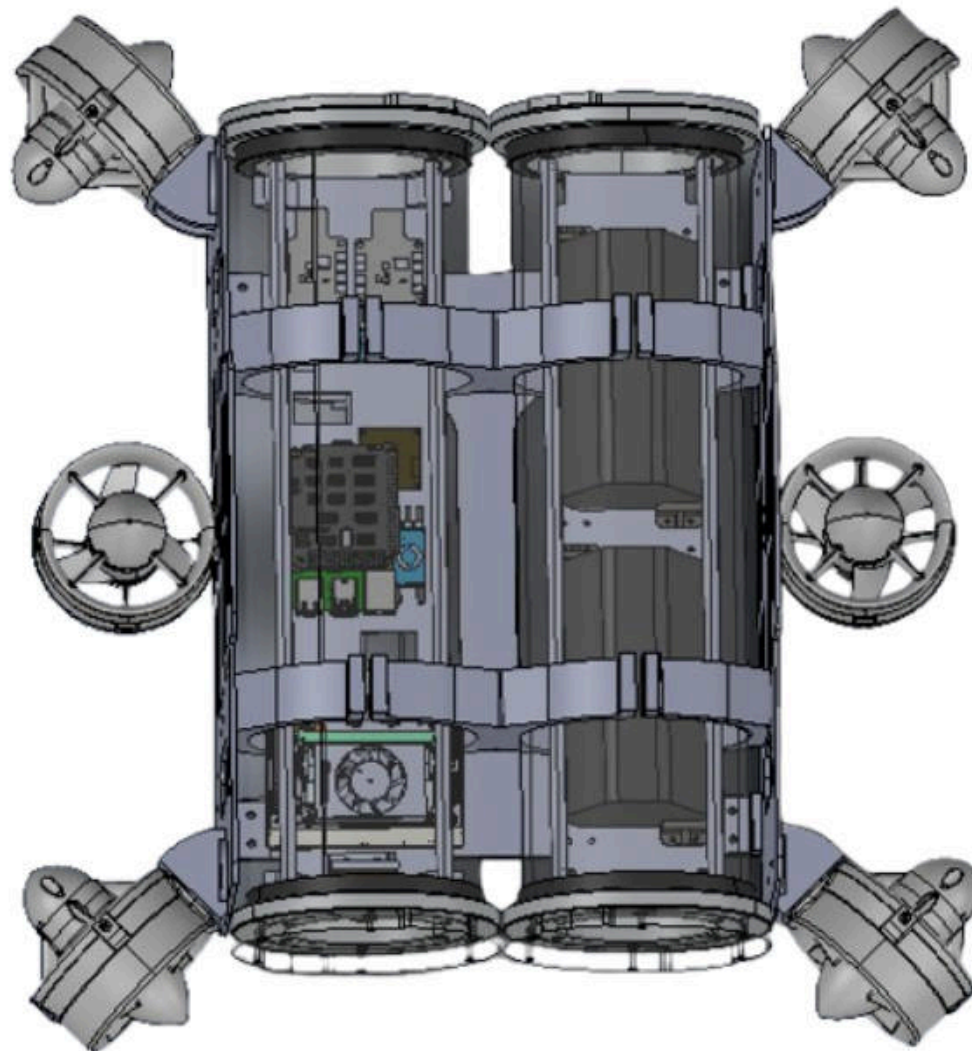
02 Object Detection

- Real Time Object Detection (Gate, bouy)

03 Manipulation

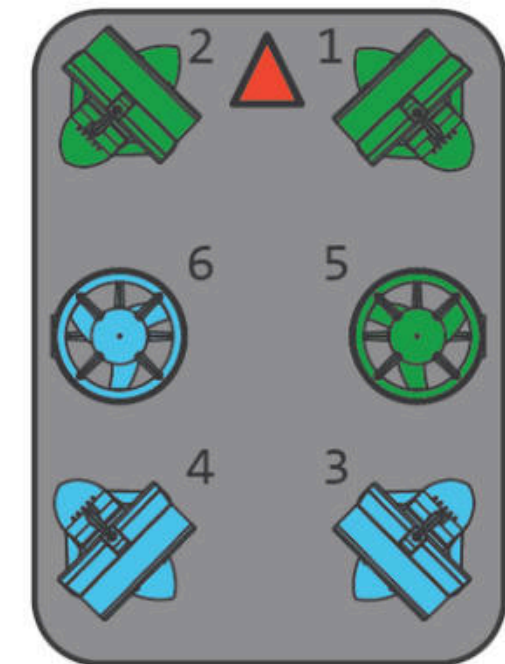
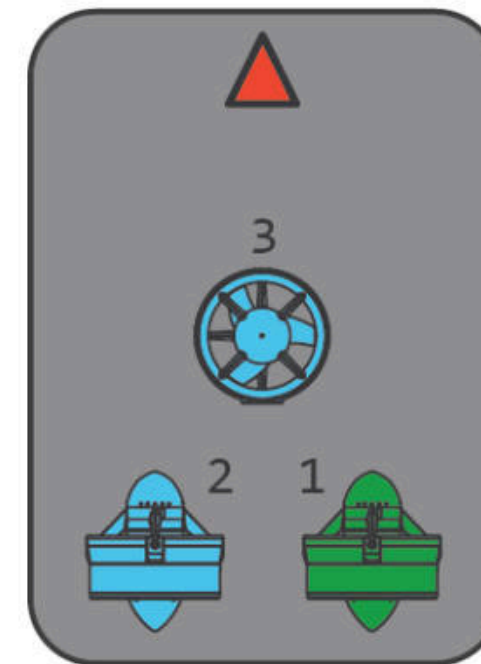
- Design and build the arm with gripper
- Pick up the object

01 AUV Navigation

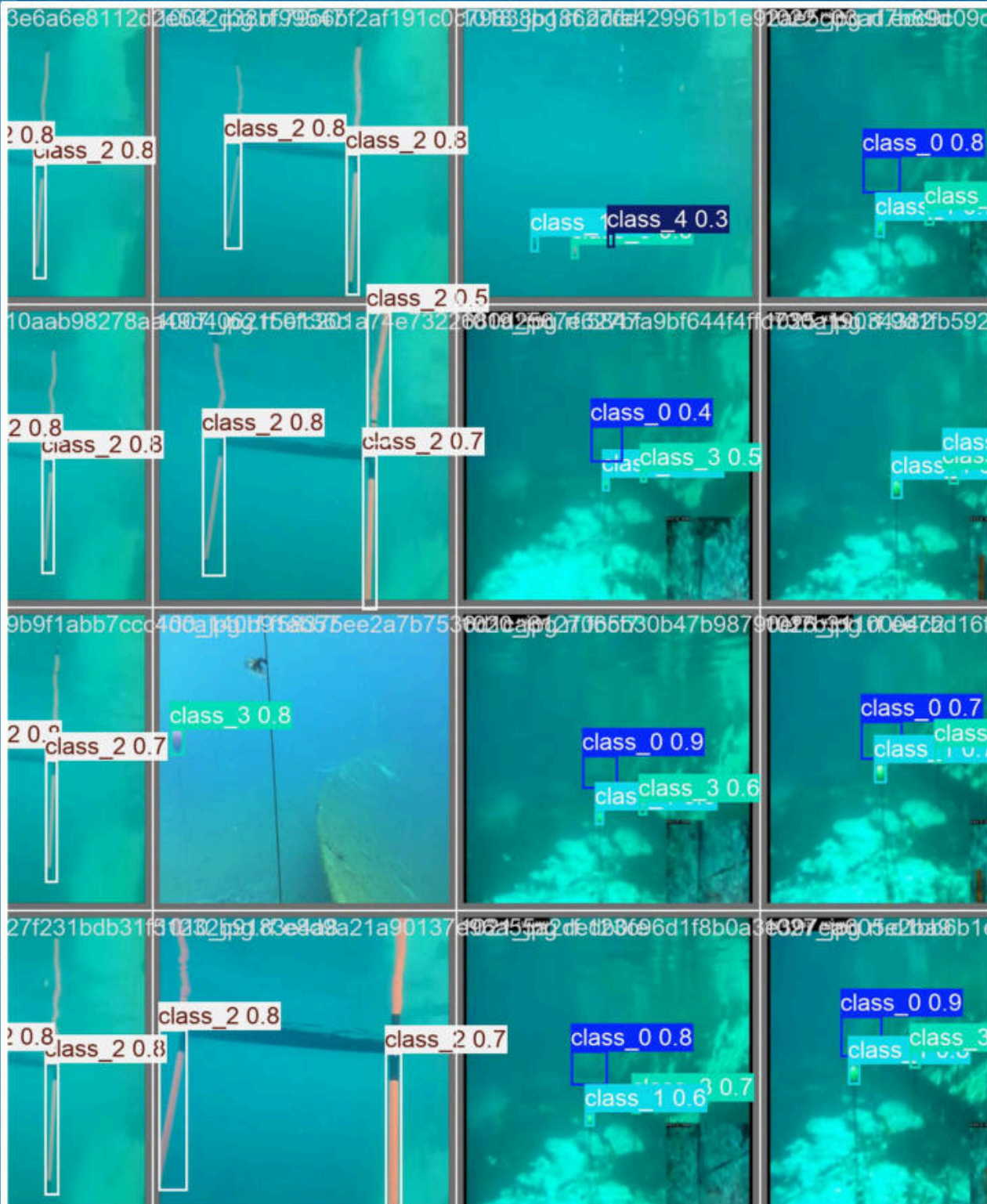


Team Offset

- Understand and adapt existing structural, electrical and mechanical components of the AUV
- Change configuration of thrusters to improve previous design :

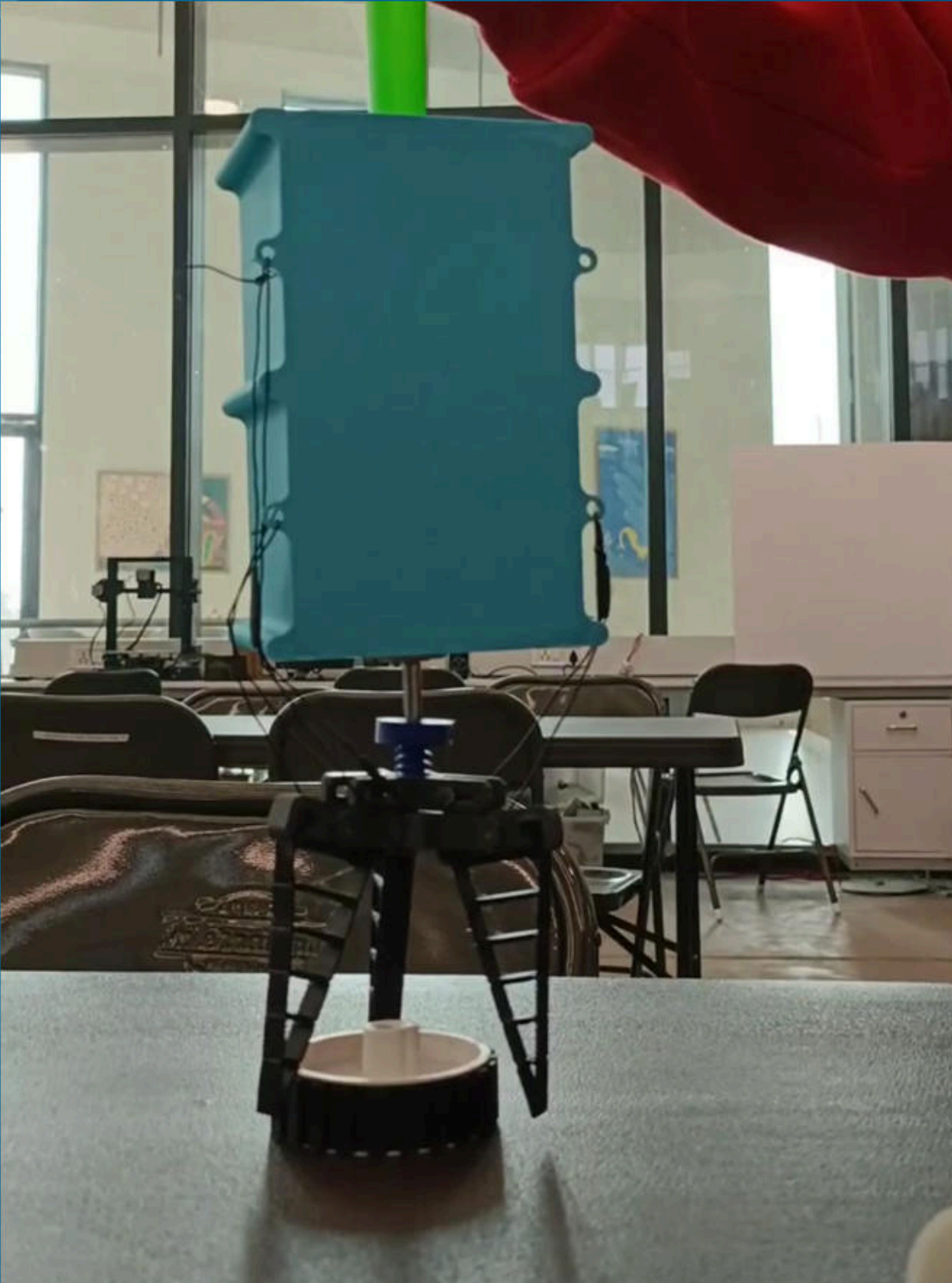


02 Object Detection

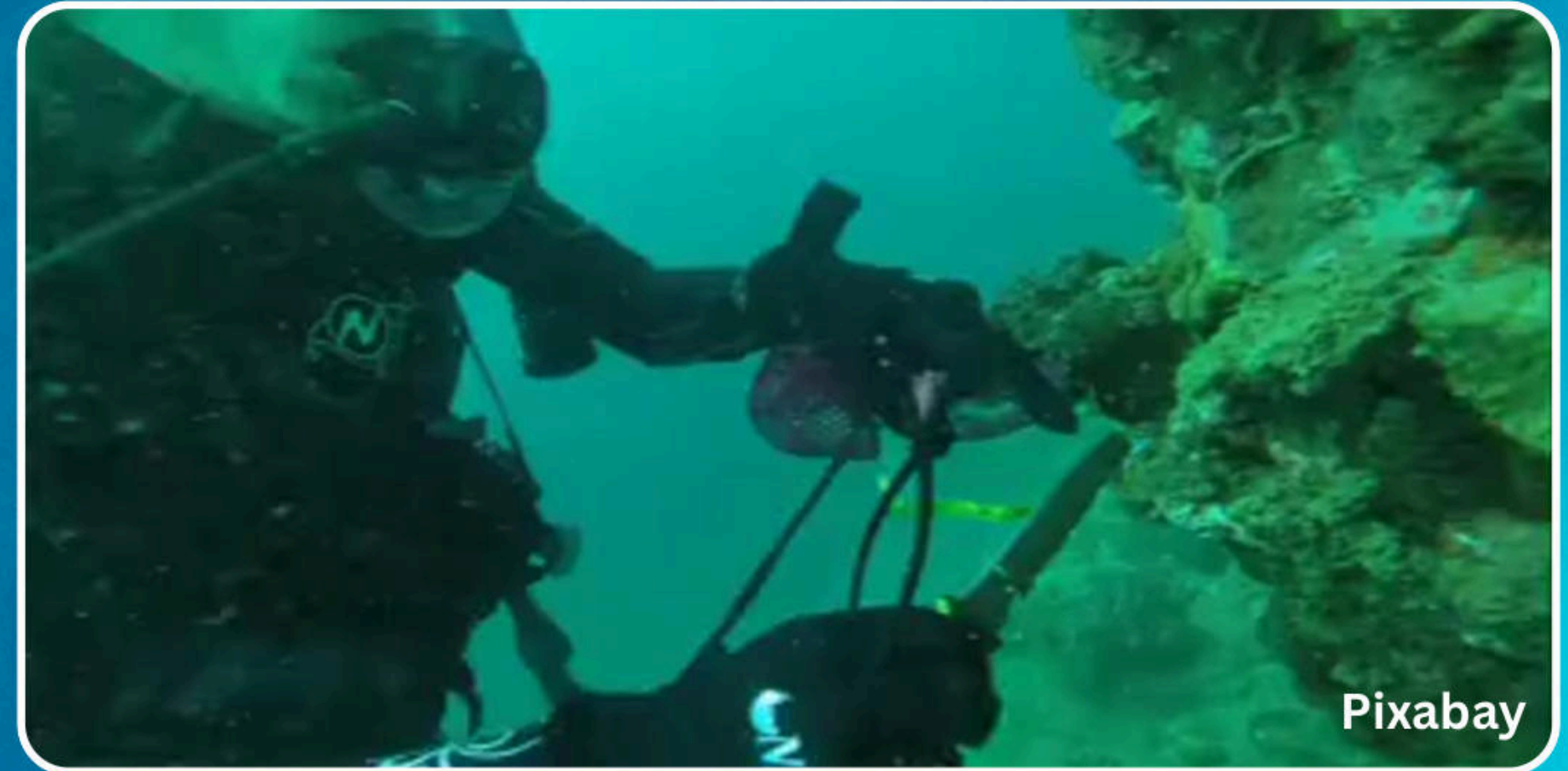
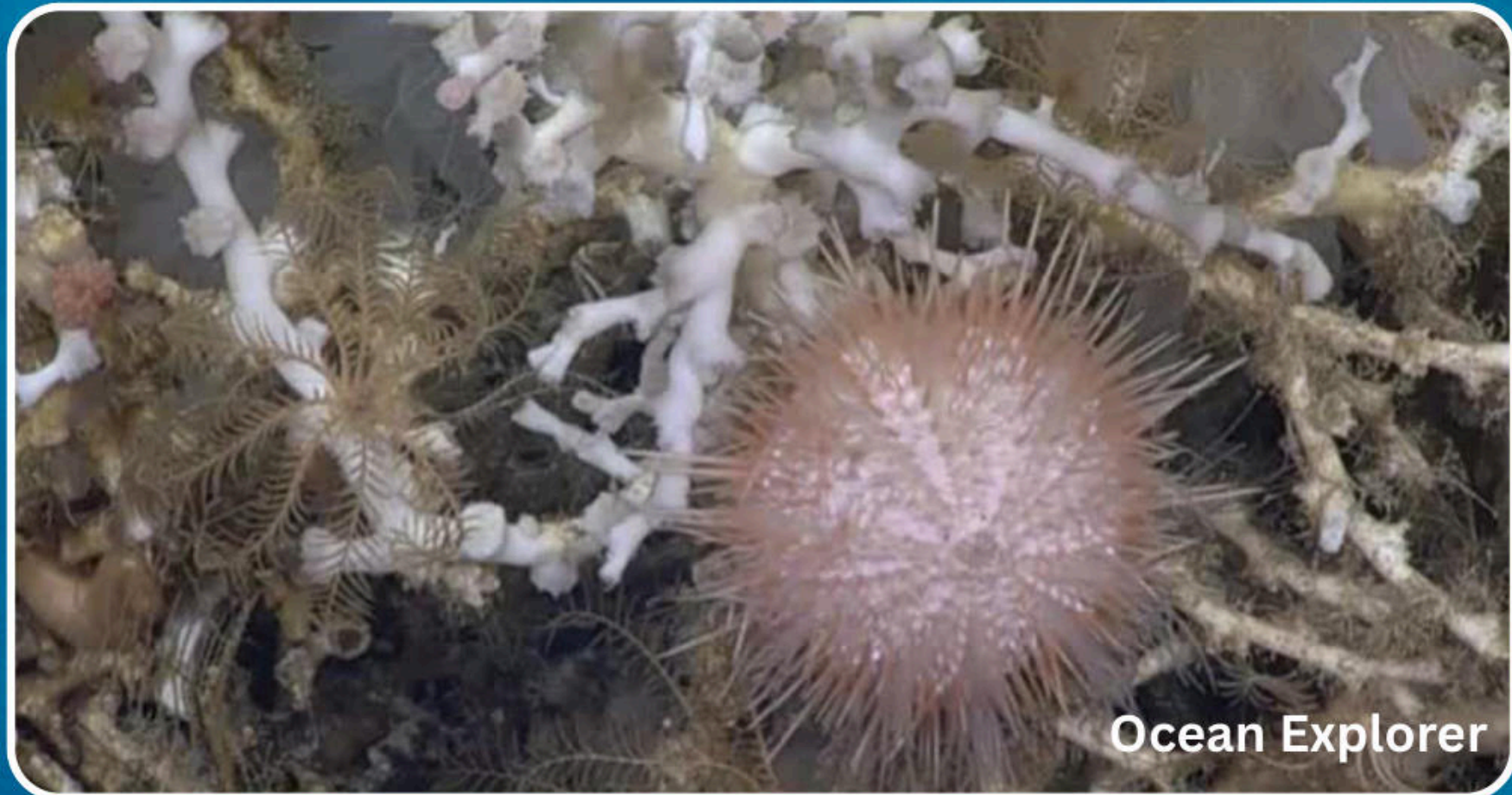
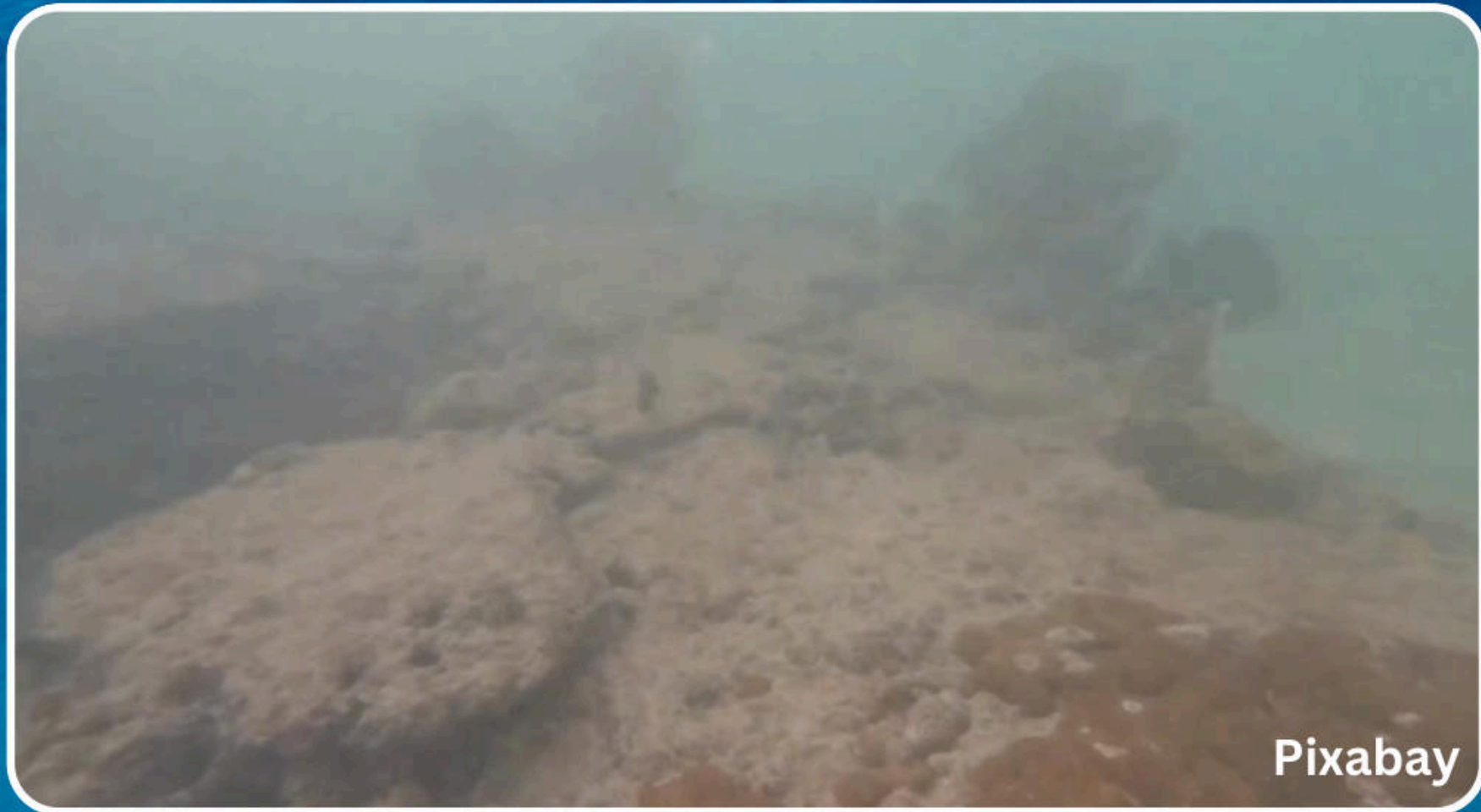


- *Created Own Underwater Image Dataset* for the task of gate and buoy detection
- Data Augmentation Pipeline: corrected white balance, hue shift, motion blur etc. to overcome turbidity
- Compared various models like **MAS-YOLOv11** and **LUIEO** to get the best performance of:
 - Precision: 78.8
 - Recall: 70.3
 - F1 Score: 74
 - mAP@50: 77.4

03 Manipulation



- **Previous manipulator:** A Cup with lid - it worked for simple pick-and-drop
- **New approach:** Shifting to a *gripper* inspired by *Fish Fin Ray*. The fingers are soft and adaptive that conform to irregular shapes
 - **Material Used** - TPU 92 Shore A in fingers for flexibility



Problem Statement

Literature Review

Modelling & Analysis

Testing

Problem Statement

Problem Statement

Design and prototype a lightweight, water-resistant underwater robotic gripper for an AUV, capable of reliably grasping diverse object types including thin, circular, delicate, slippery, and rigid objects

Literature Review

A Taxonomy of Underwater Manipulative Actions

Main issues:

- Squash soft samples (corals, sponges) on full closure
- Poor force control → relies on pilot experience
- Slow closure → fast organisms escape
- No force/tactile feedback
- Slow/risky tool changes in current

Manipulative Actions for sampling a marine organism:

- Detachment
- Collection
- Storage



National Geographic

Design Requirements for Underwater Grippers

Task-Related:

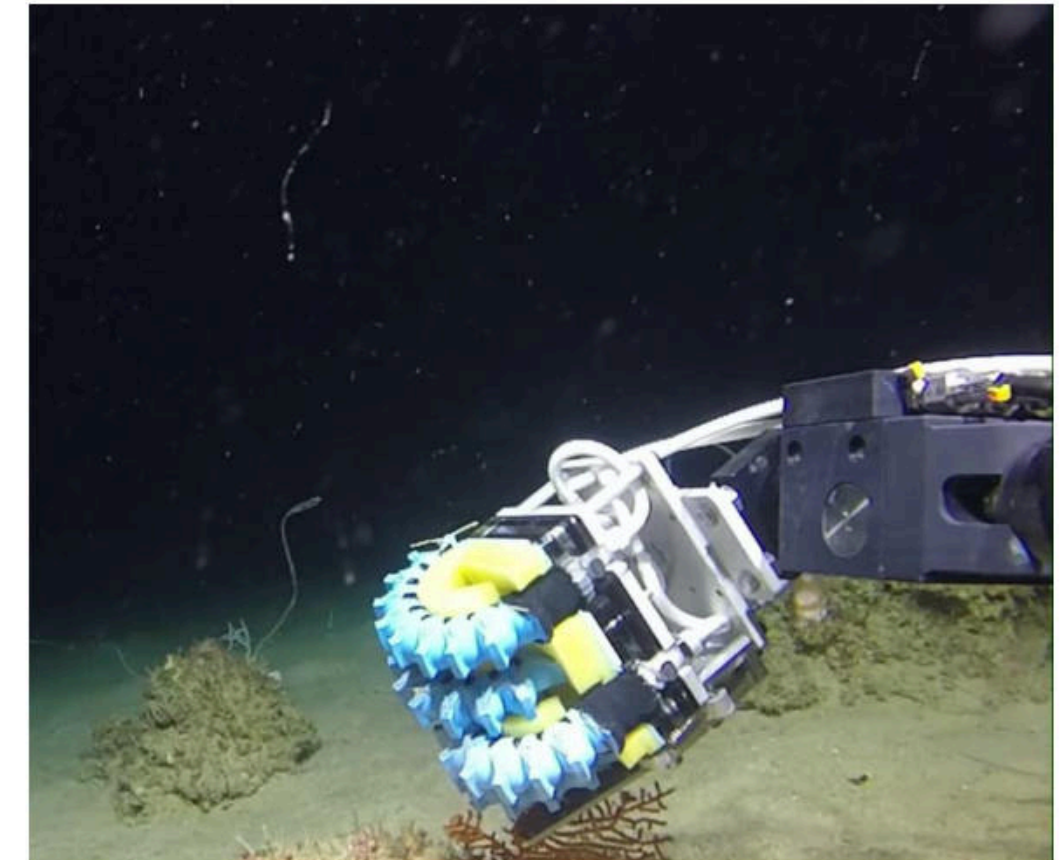
- Precise tuning/control of gripping force
- Fast closure/retraction to capture specimens before they retract or escape
- Rigid, secure holding during transport/storage.
- Prehensile/enveloping grasp options for fragile samples

Operational/Functional:

- Force/tactile feedback for better control

Environmental/Deep-Sea Specific:

- Full water/pressure resistance
- Near-neutral buoyancy
- Minimal volume change under pressure (to maintain buoyancy control)



National Geographic

Existing Underwater Grippers: Mechanism

3 Main Parts of a Gripper:

1. **Grippers:** 4 classes of manipulator grippers:

- Parallel fingers,
- opposed/intermeshed fingers,
- grabber claw and
- cage claw

2. **Sampling Tools (manipulated by gripper)**

3. **Storage System:** Suction, scoops, corers (sediment collection), traps and Niskin bottles (collect water samples, syringe-based systems)

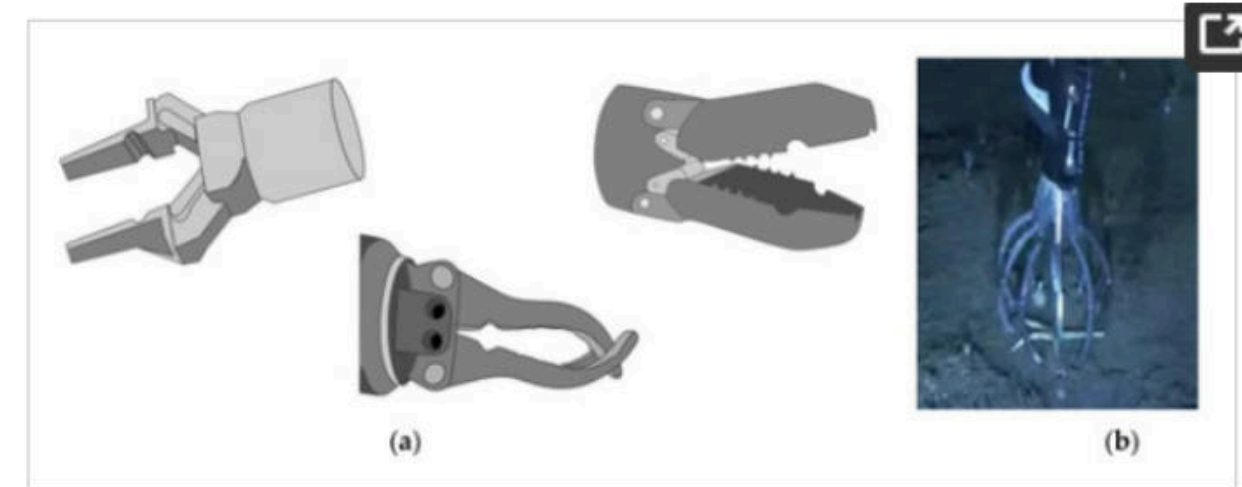


Figure 1. (a) From left to right, examples of parallel, intermeshed and grab claw. (b) Example of cage gripper: gripper of the ROV LIROPUS 2000. (Reprinted with permission from ref. [22]. © 2022 IEO).

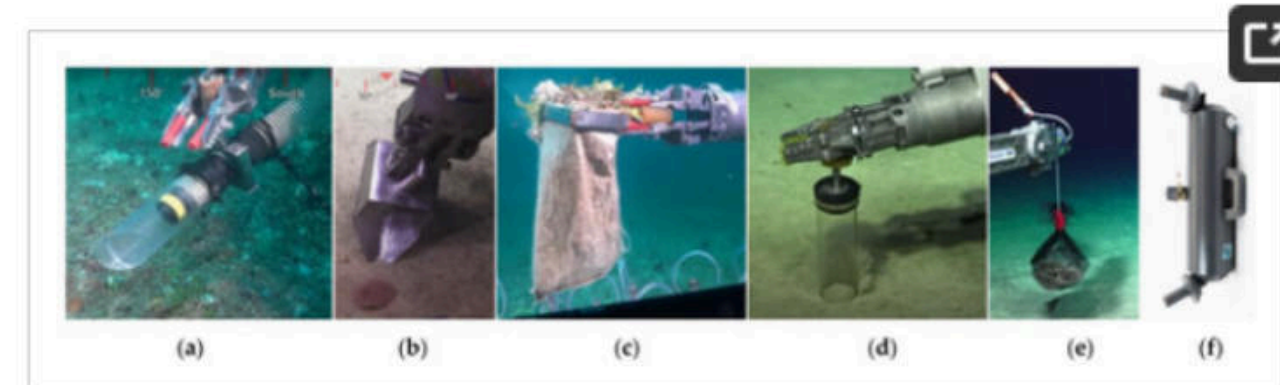


Figure 2. ROV collection tools: (a) suction sampler; (b) scoop; (c) scoop net; (d) corer; (e) trap; (f) Niskin bottle. ((a–e) Reprinted with permission from ref. [23,24,25,26,27]. © 2022–2021 Schmidt Ocean Institute. (f) Reprinted with permission from ref. [28]. © 2022-2022 General Oceanics).

Existing Underwater Grippers: Trends

Bio inspiration for task-specific designs and a shift towards **soft grippers**

- **Advantages:**

- Tolerance to positioning errors
- Reduced damage to organisms

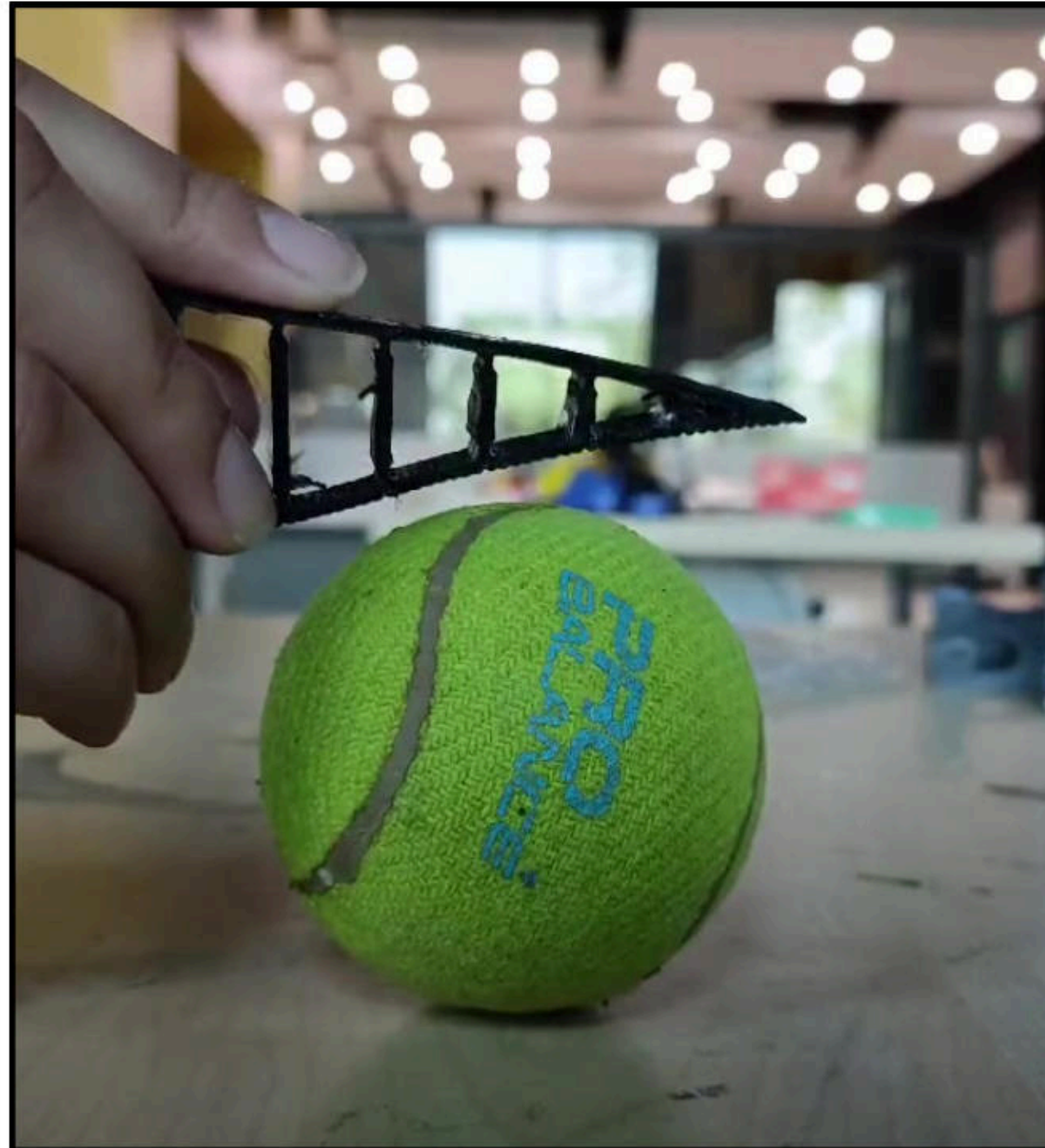
- **Limitations:**

- Lower mechanical strength
- Risk of tearing under cyclic pressurisation

- **Ongoing improvements include:**

- Fiber reinforcement
- Better interface design between soft and rigid parts

Current Design



Fin Ray Finger

- Inspired by Fish ray fins
- Grips by conforming to the object's shape
- 3D-printed TPU

This Sem:

- Explore the number of fingers
- Test different finger infill structures
- Add wedges

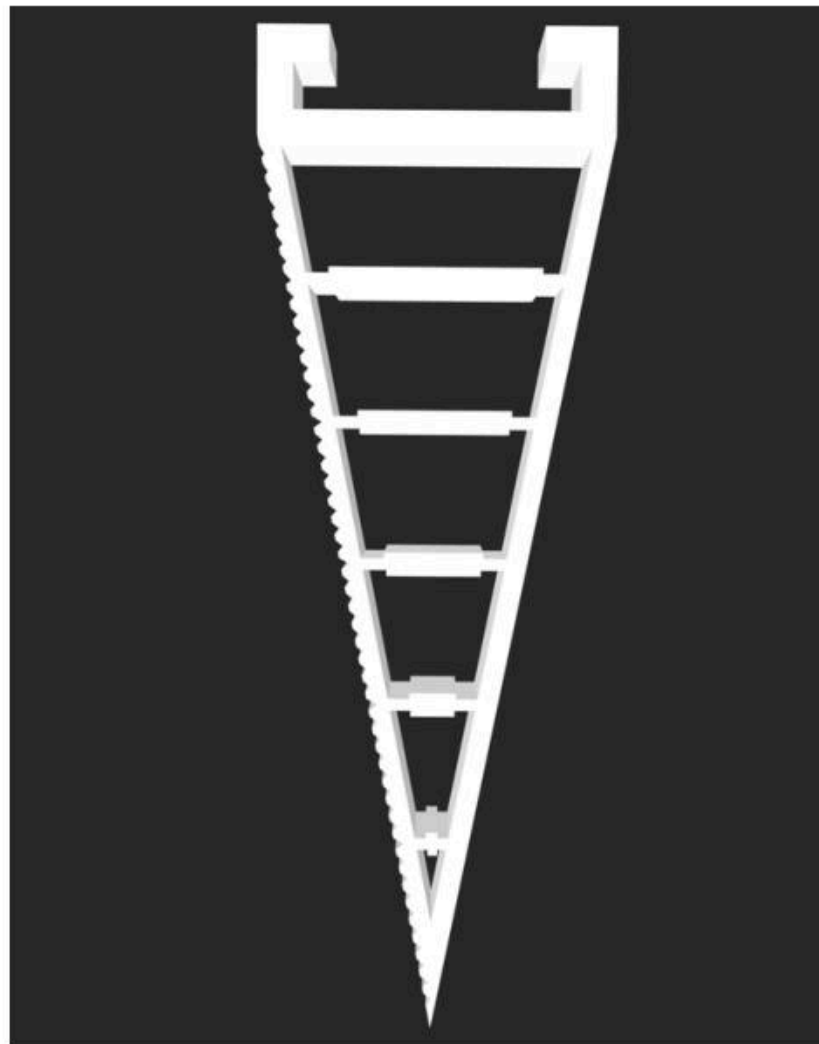
Modelling & Analysis

01

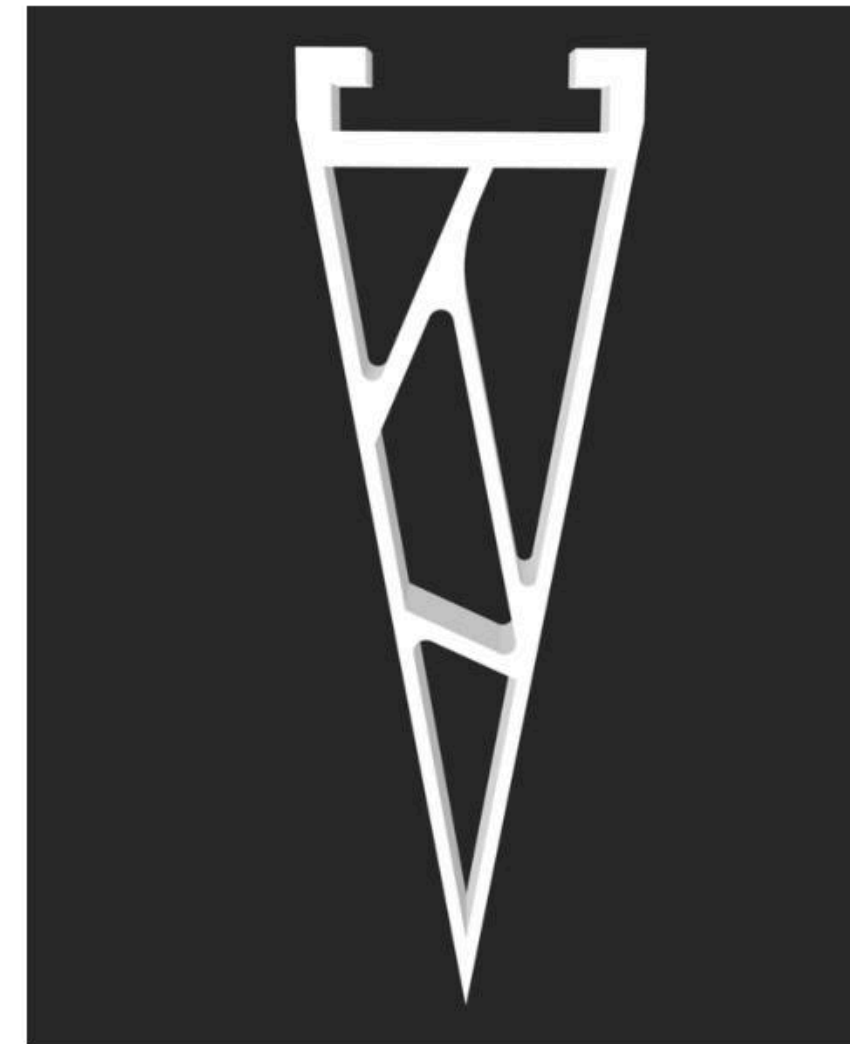
Fin Ray Finger

01 Internal Fill Geometry

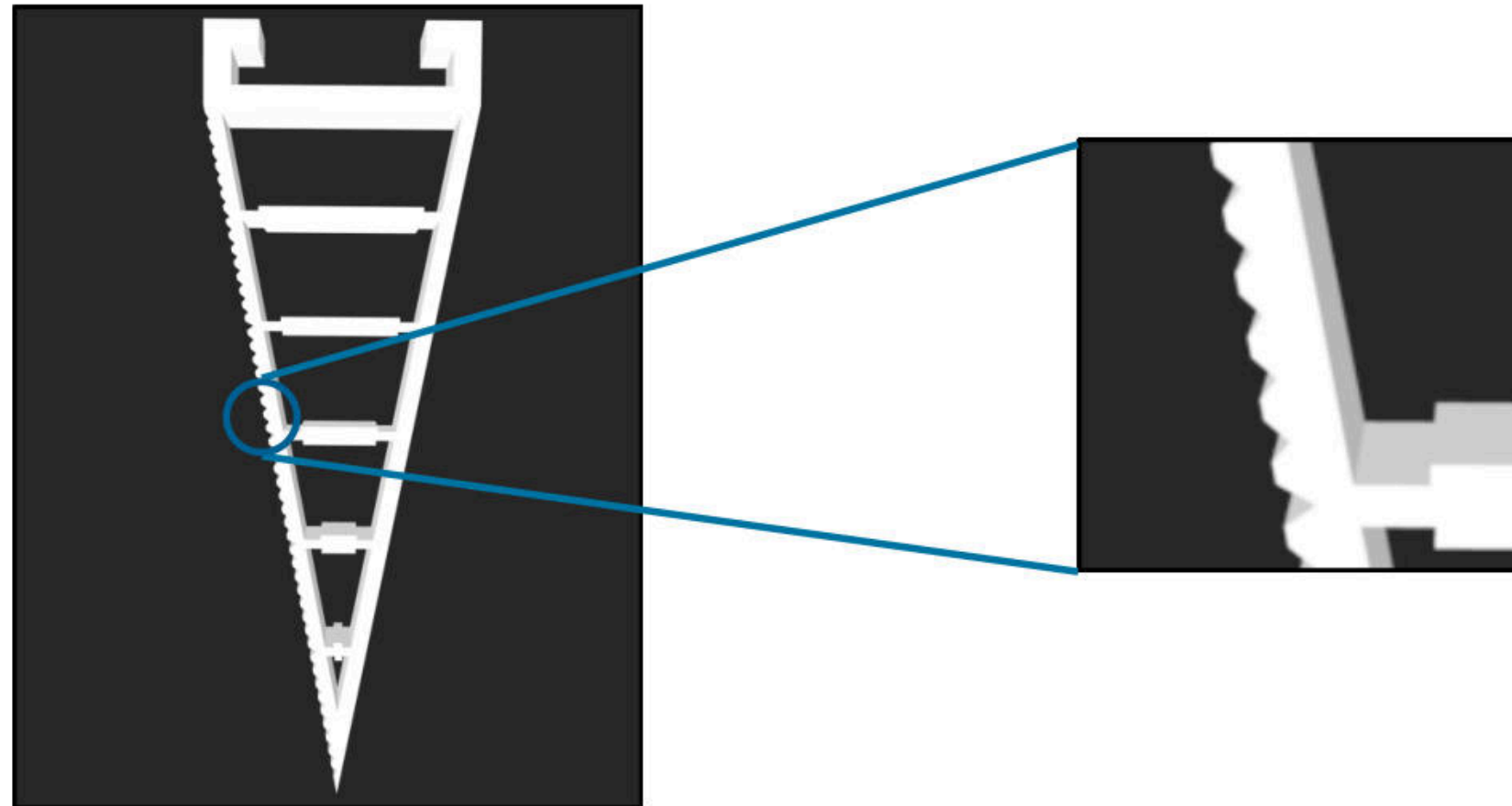
Standard Fin Ray Design - Straight Struts



Alternate Design - Inclined Struts



02 Microwedge Structure



Microwedge profile along one side of the finger body to increase friction and grip strength on contact surfaces

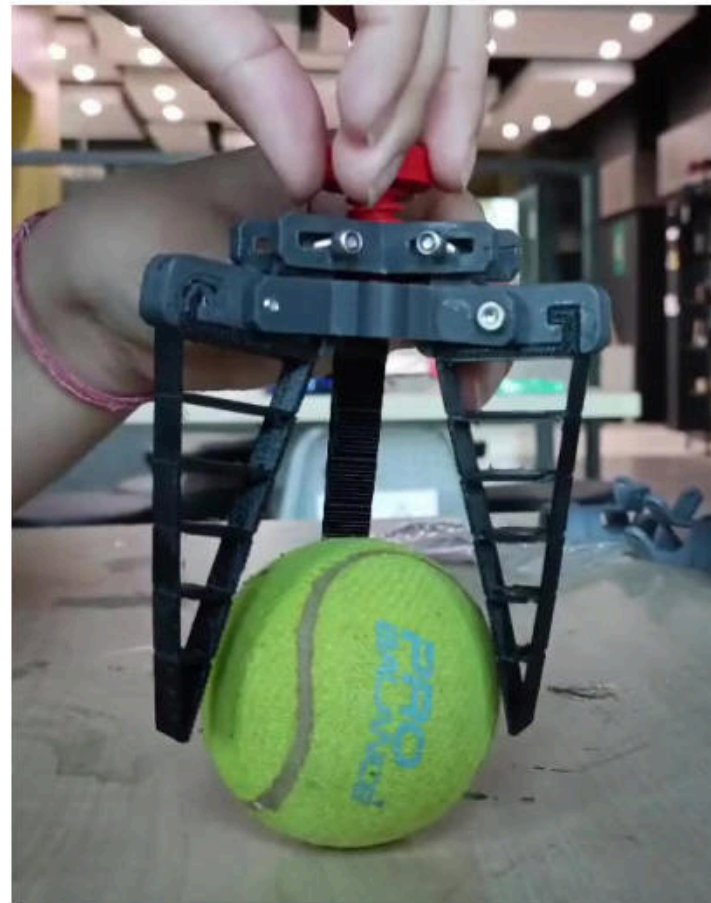
02

Actuation System

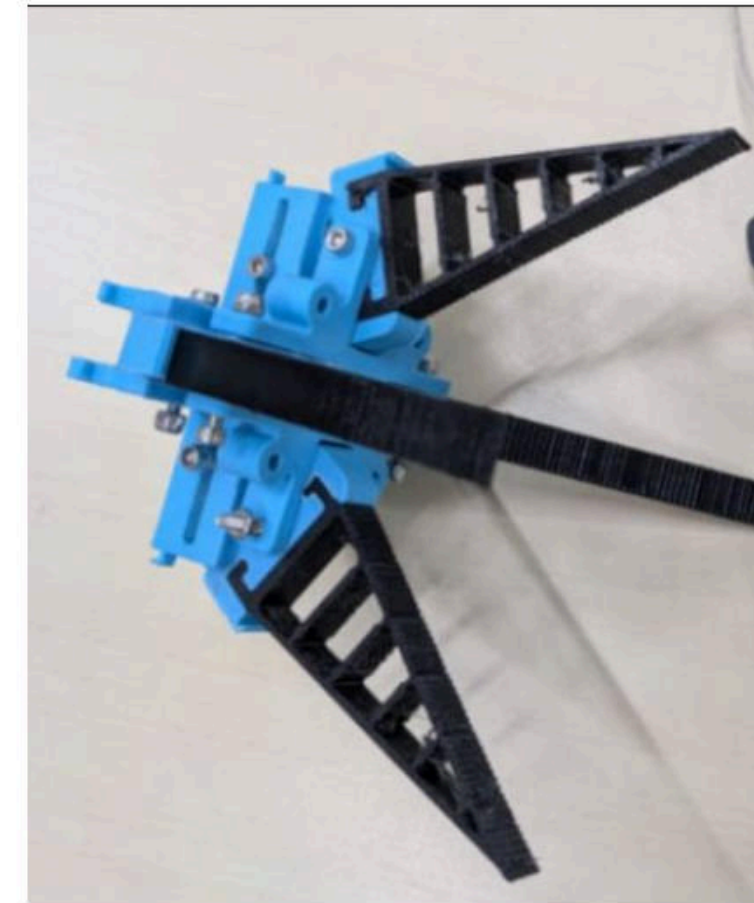
Design 1 →

Worm-Follower Actuation

01 3-finger to 4-finger Configuration



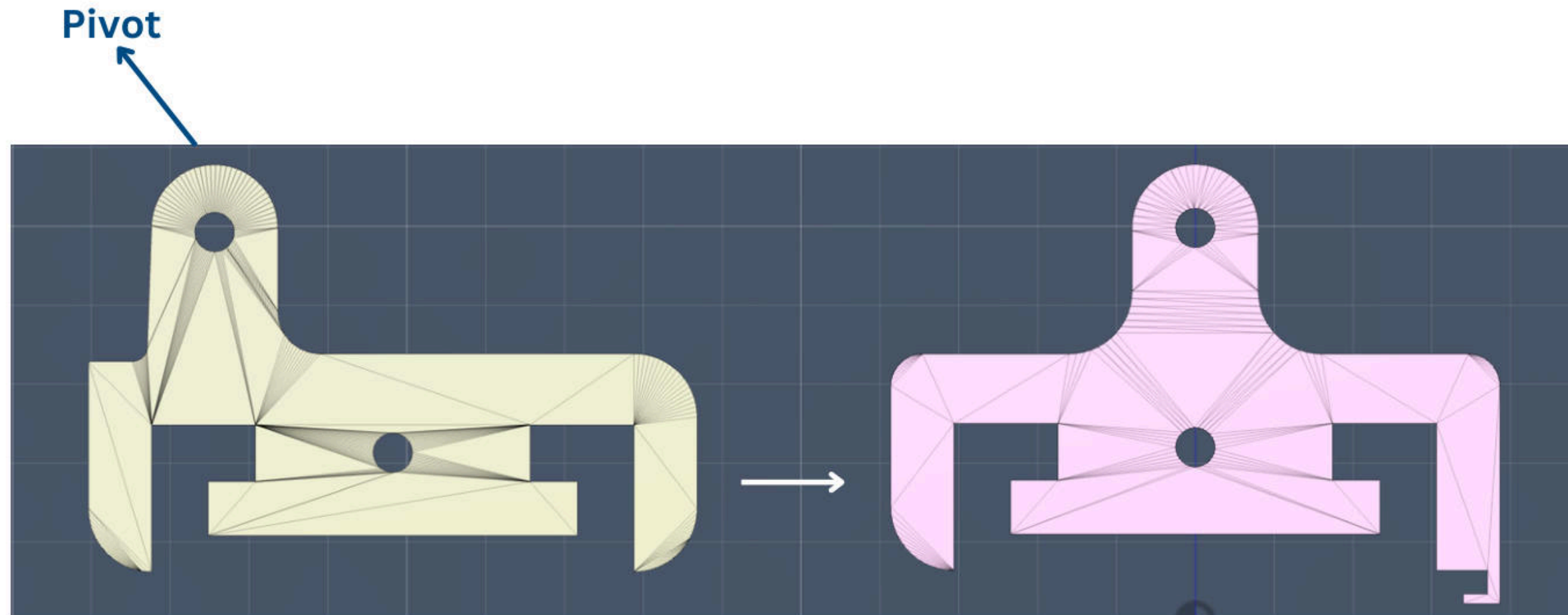
3 finger configuration



4 finger configuration

Inspired By → "LAD Robotics"

02 Pivot Relocation

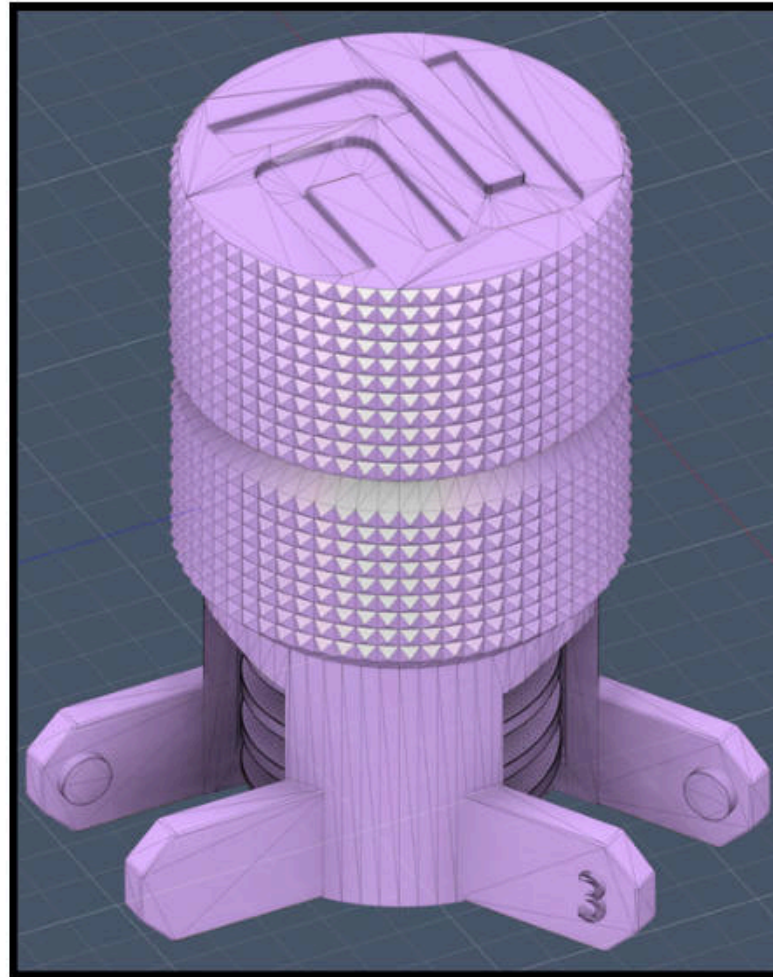


- Changed the pivot point in the clamp to increase the opening range
- The worm follower actuation was not working

Design 2 →

Rack-and-Pinion Actuation

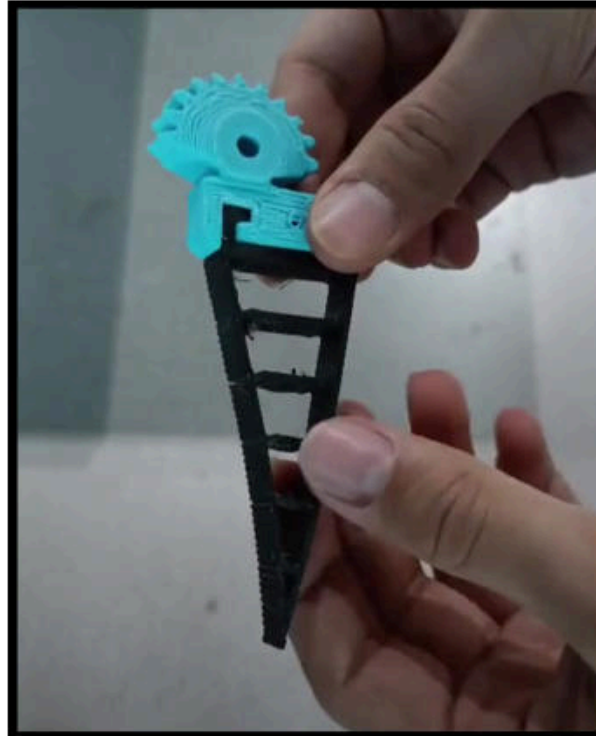
01 3-finger to 4-finger configuration



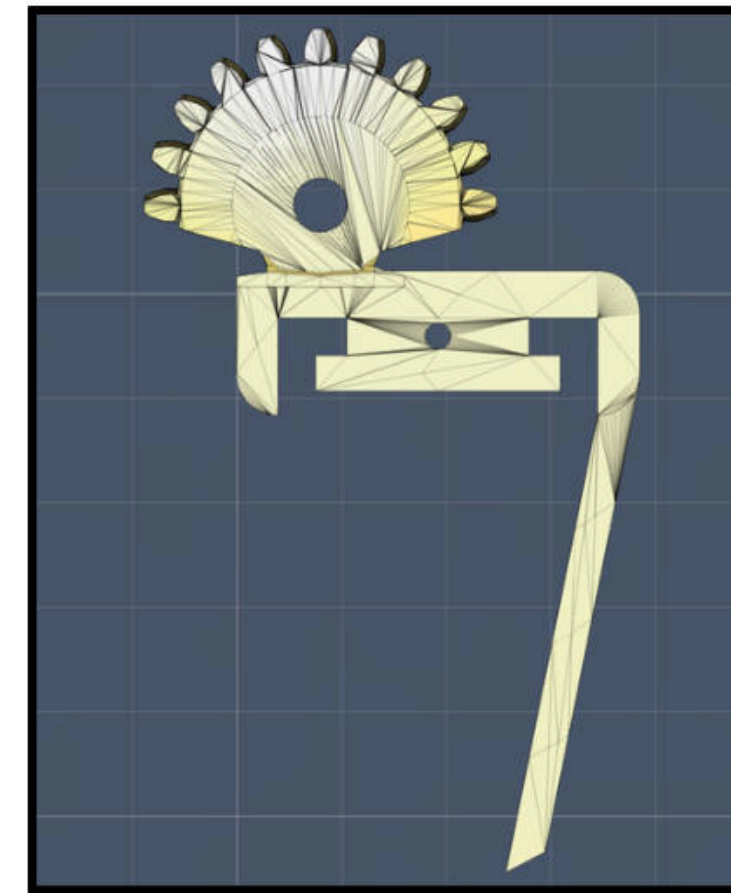
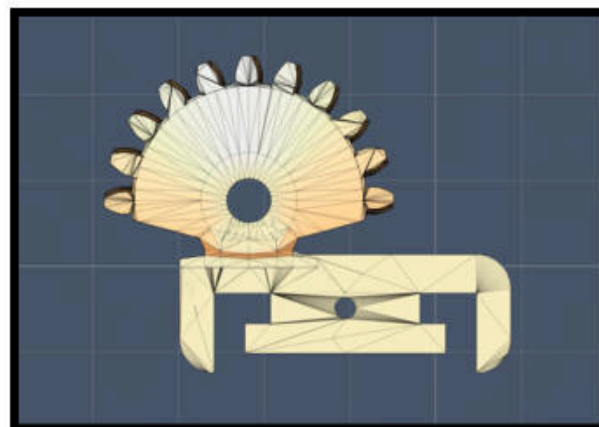
- As the pinion rotates, all fingers are driven simultaneously
- Wide opening and closing range

Inspired By → "PrintChallenge"

02 Clamp Redesign

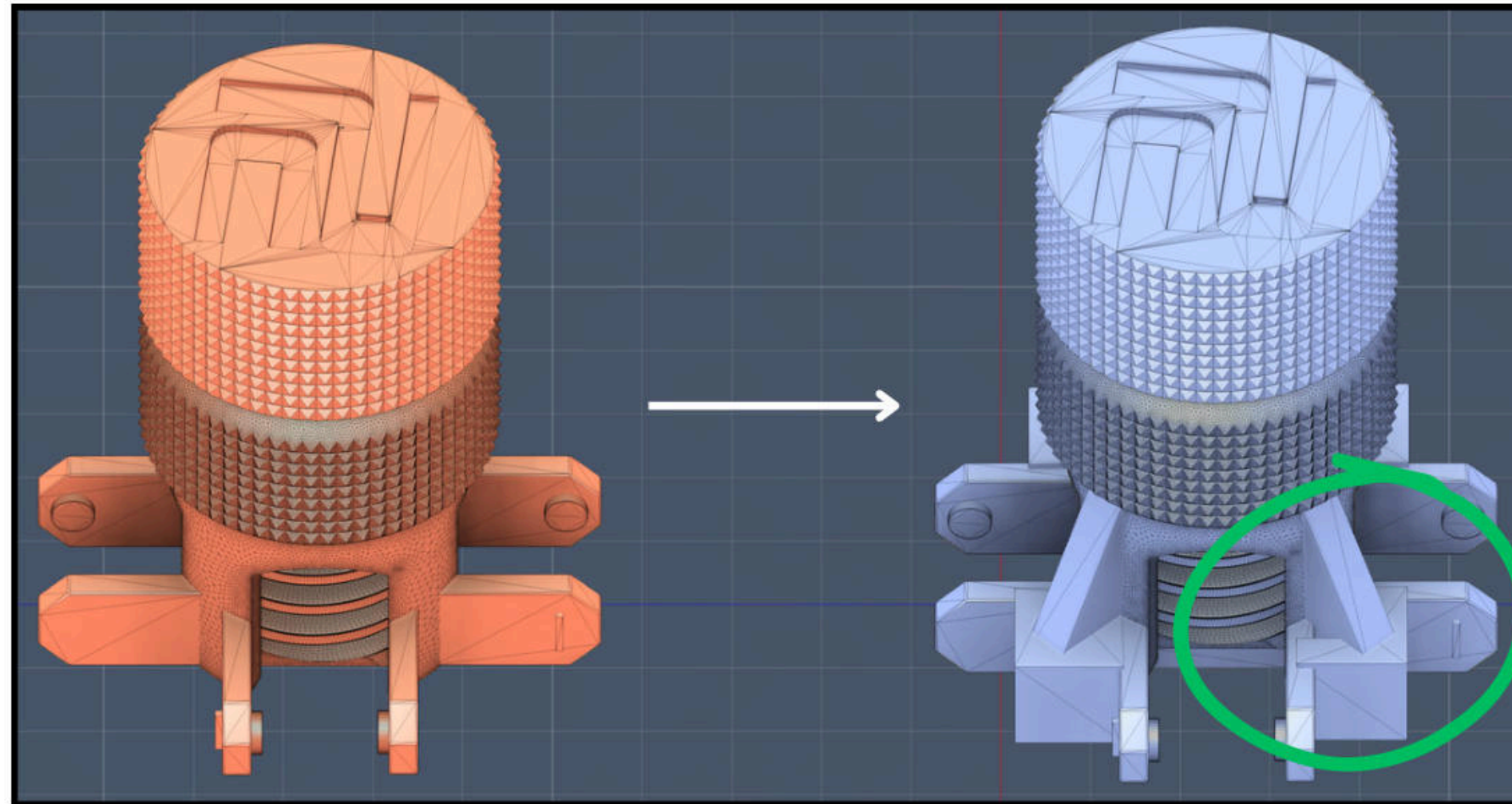


Lateral Bend



Redesigned the clamp to add extended support
eliminating the bend during closure

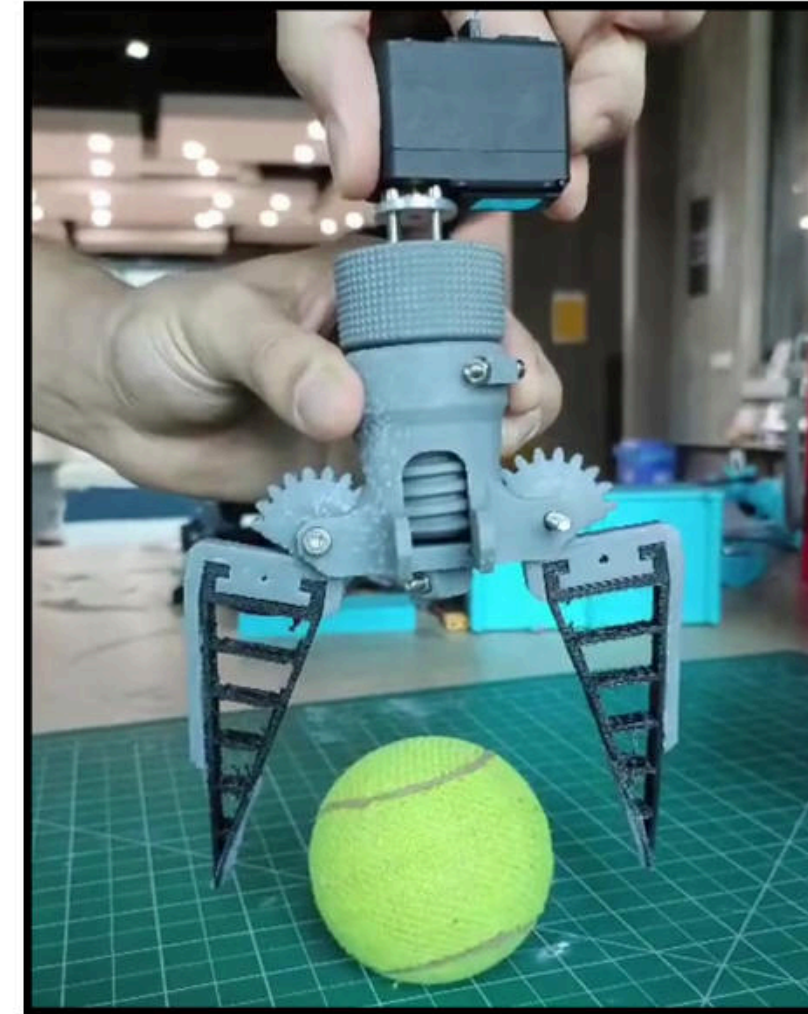
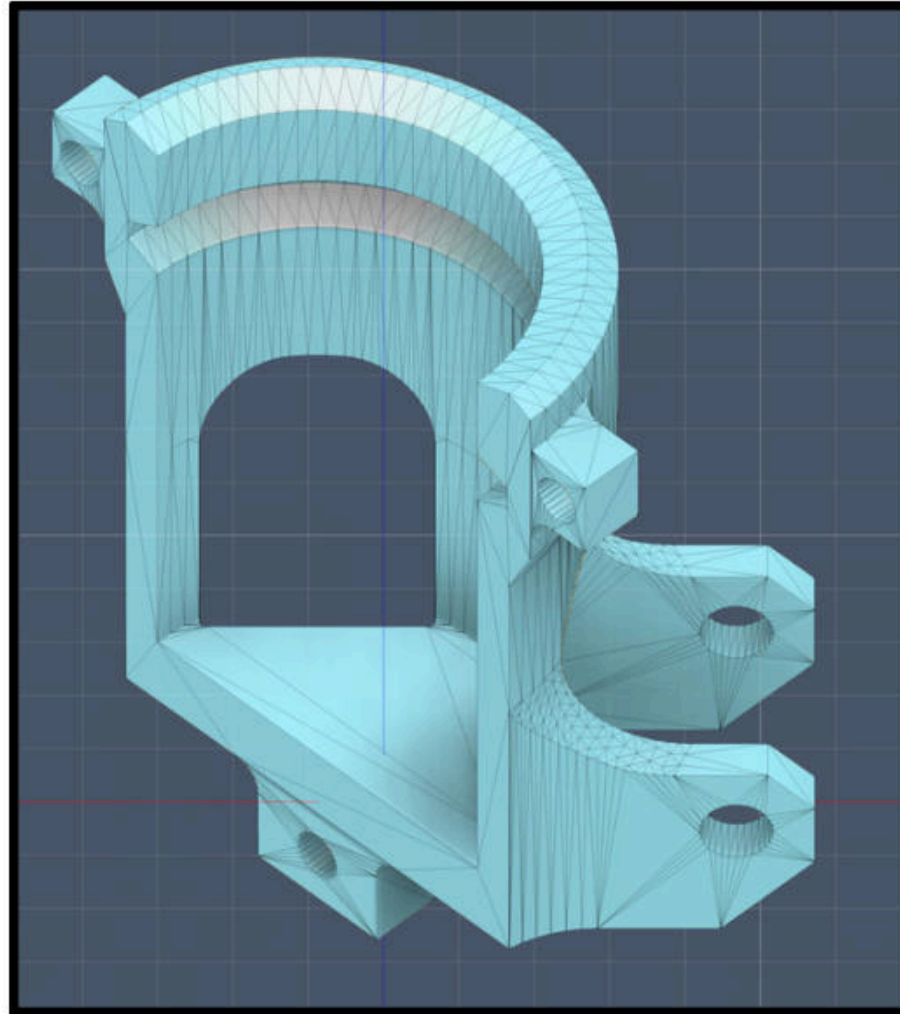
03 Added Support for Structural Strength



X Structural Failure

✓ Added Support

04 Split the Assembly



For ease in SLA printing!

05 Aluminium Design



In Progress

Testing

TESTING METHODOLOGY AND METRICS

To test our various gripper designs, we referred to the performance metrics derived for robotic arms and grippers by the National Institute of Standards and Technology

01

Performance metrics: Slip
Resistance, and Grasp
Strength

02

Split cylinder used for
evaluations

03

Yale-CMU-Berkeley Object
set

Falco, J., Van Wyk, K., & Messina, E. (October 2018). Performance Metrics and Test Methods for Robotic Hands (DRAFT NIST Special Publication 1227). National Institute of Standards and Technology, U.S. Department of Commerce
CB Benchmarks. (n.d.). Results & records. YCB Benchmarks – Object and Model Set. <https://www.ycbbenchmarks.com/world-records/>

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Testing

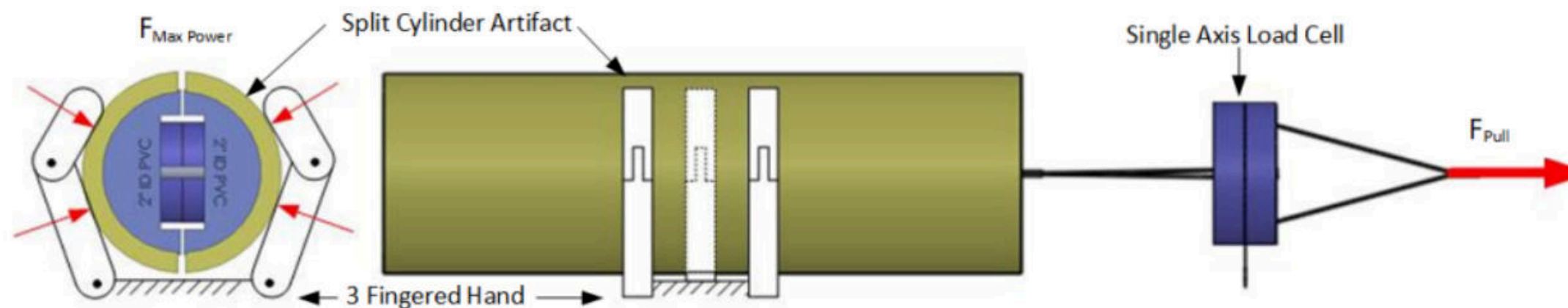
Slip Resistance

What is it?

- Tests a hand's ability to resist slipping by measuring the maximum pull force it can withstand under a full-force wrap grasp

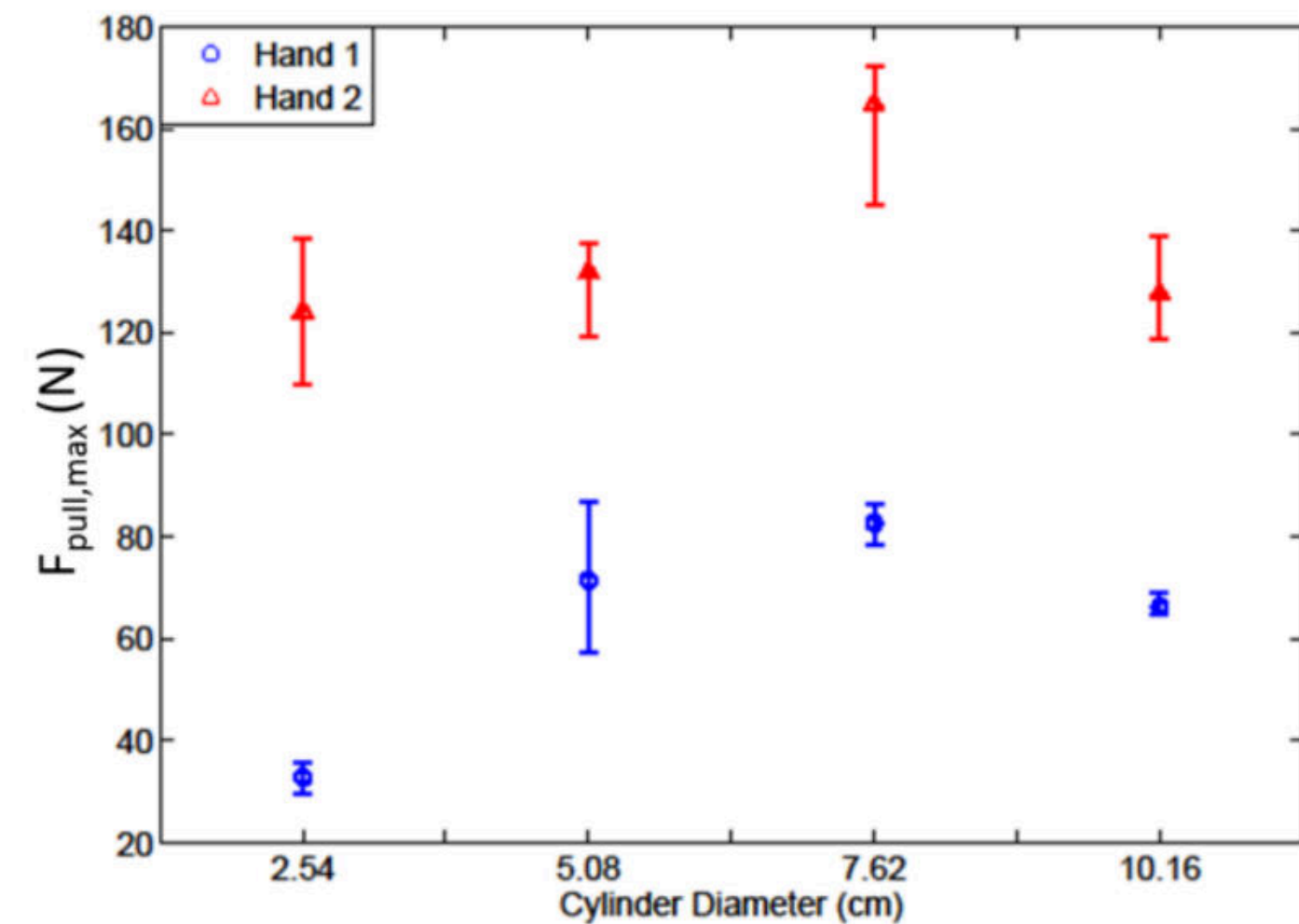
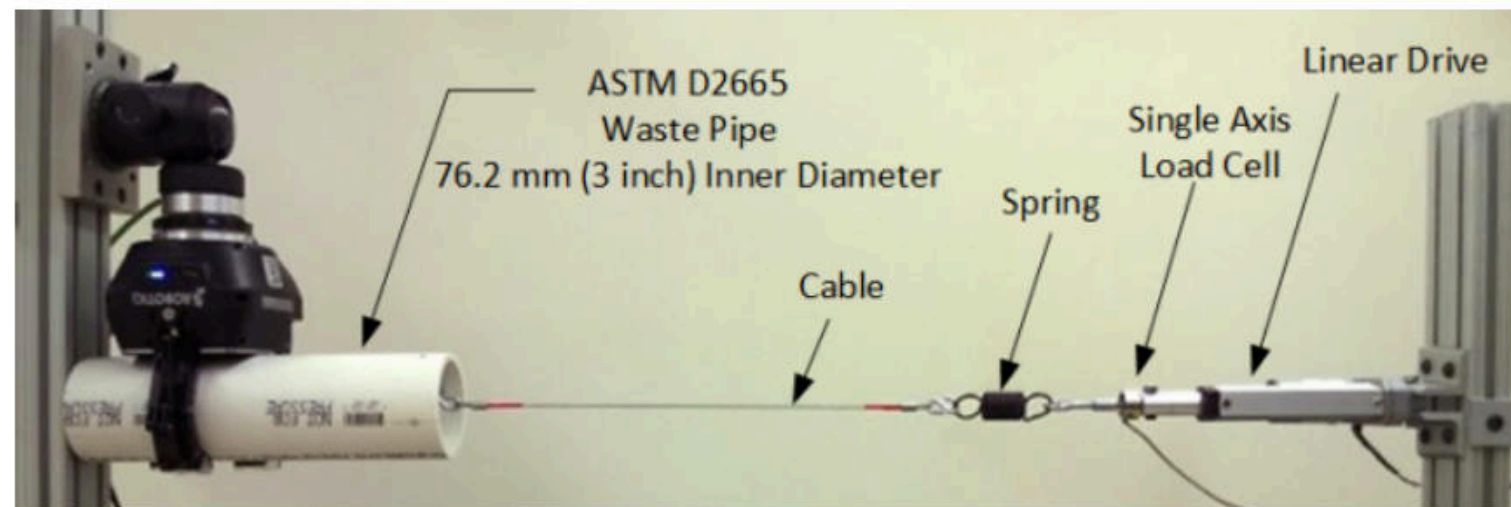
Evaluation

- Cylindrical artifact grasped at maximum power, and a controlled, increasing axial force is applied to the object until "gross slipping" is visually confirmed
- The maximum pull force is taken note of



Falco, J., Hemphill, D., Kimble, K., Messina, E., Norton, A., Ropelato, R., & Yanco, H. (2020). Benchmarking Protocols for Evaluating Grasp Strength, Grasp Cycle Time, Finger Strength, and Finger Repeatability of Robot End-effectors. IEEE Robotics and Automation Letters, 5(2). This author manuscript was made available in PubMed Central (PMC) on November 17, 2020.

Slip Resistance



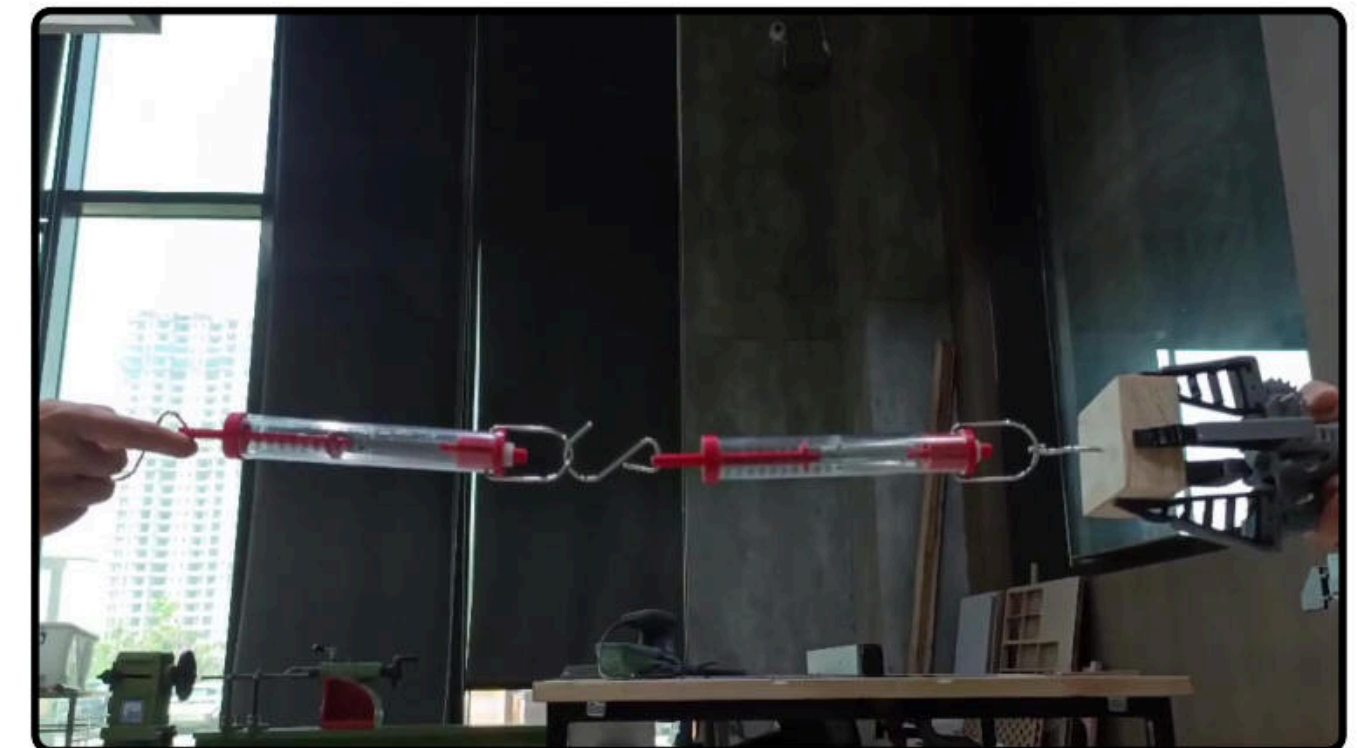
Falco, J., Hemphill, D., Kimble, K., Messina, E., Norton, A., Ropelato, R., & Yanco, H. (2020). Benchmarking Protocols for Evaluating Grasp Strength, Grasp Cycle Time, Finger Strength, and Finger Repeatability of Robot End-effectors. IEEE Robotics and Automation Letters, 5(2). This author manuscript was made available in PubMed Central (PMC) on November 17, 2020.

Test Setup & Results

Gripper	Object	Average Slip Resistance
4 Finger	Bottle	1.93 N
4 Finger	Wooden Cube	1.77 N
2 Finger	Wooden Cube	0.70 N
2 Finger	Bottle	1.08 N

4 fingers configuration performed better than 2 fingers configuration

Test Setup



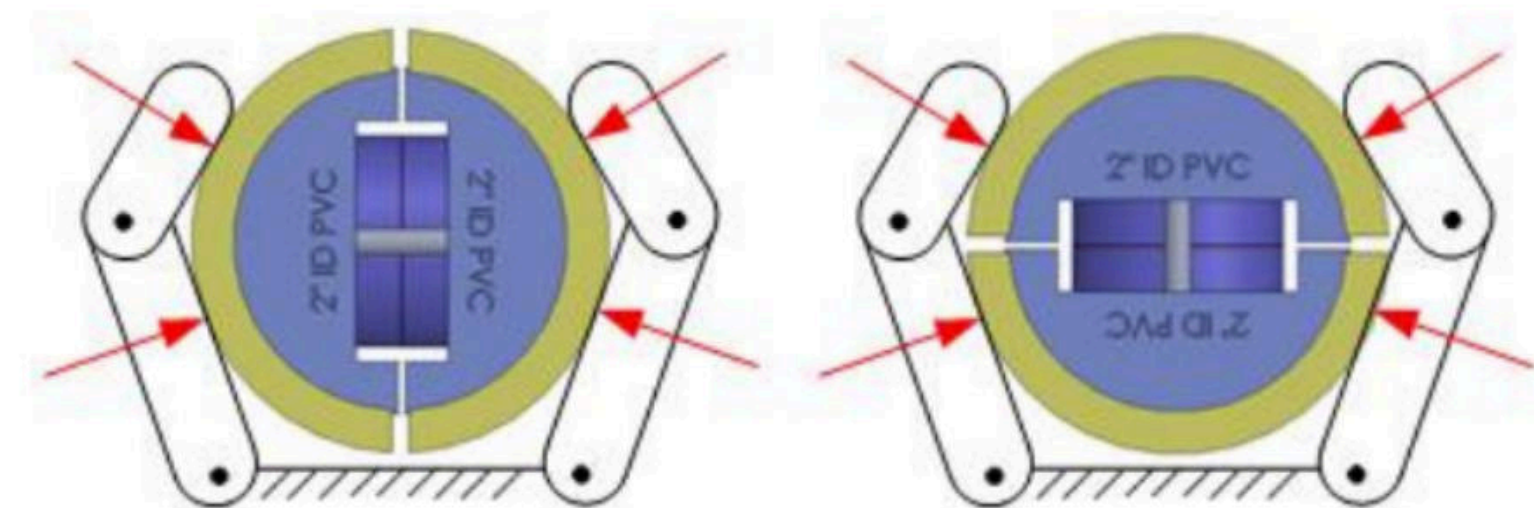
Grasp Strength

What is it?

- Evaluates the maximum force a robotic hand can impose on an object,
- provides insights into payload capabilities

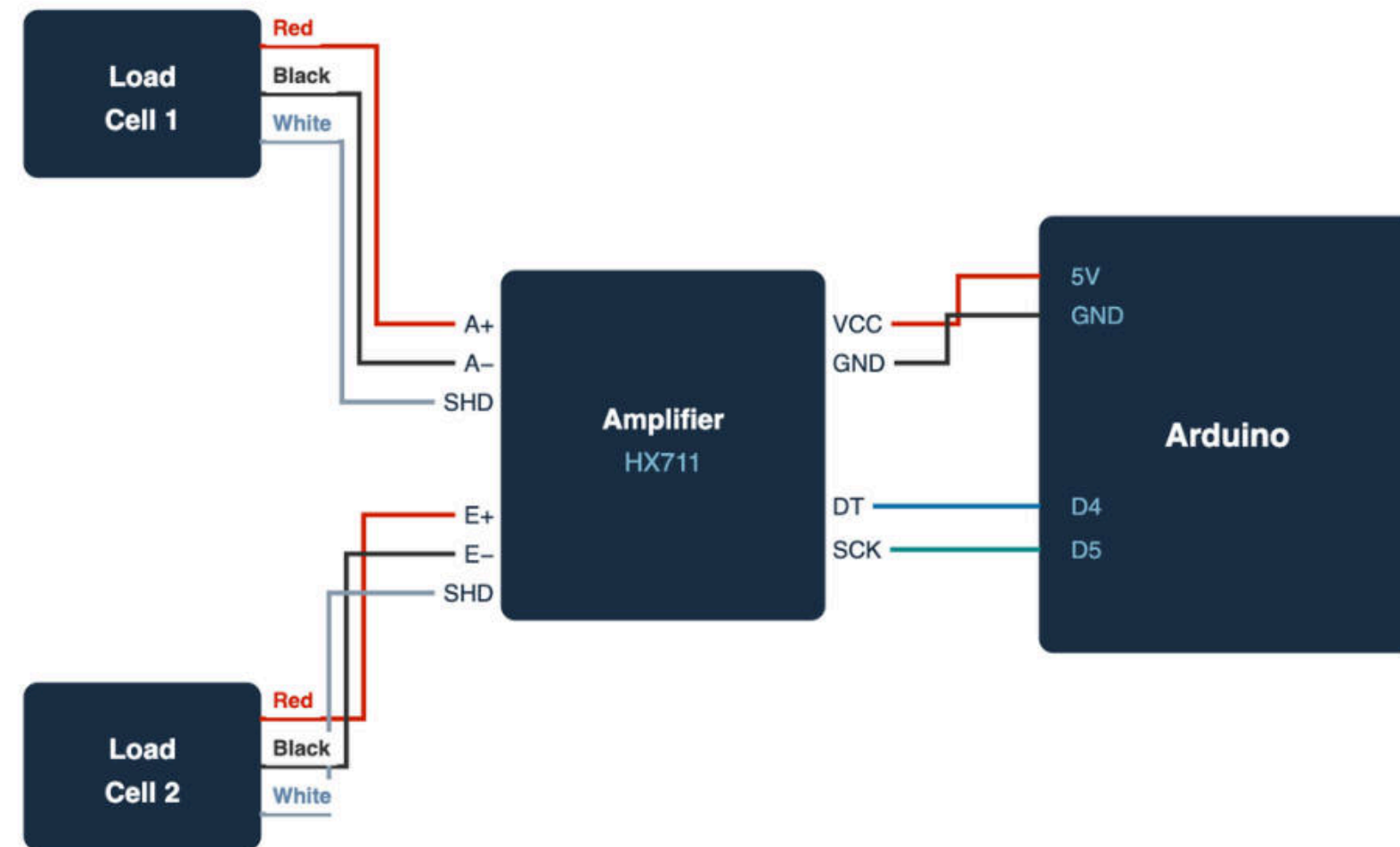
Evaluation

- Materials: split cylinder is necessary.
- Gripper holds the maximum force it can exert for 5 seconds, then retracts to open fully
- Process repeated for 32 cycles

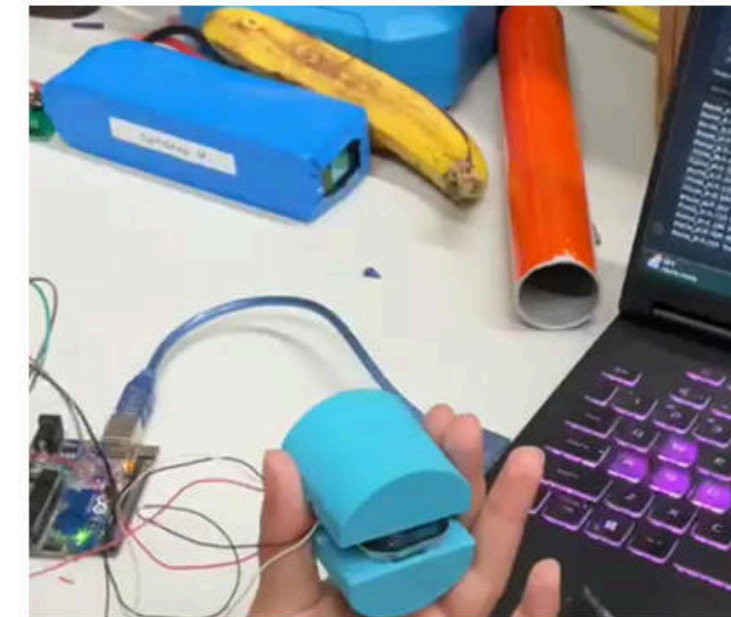


Falco, J., Hemphill, D., Kimble, K., Messina, E., Norton, A., Ropelato, R., & Yanco, H. (2020). Benchmarking Protocols for Evaluating Grasp Strength, Grasp Cycle Time, Finger Strength, and Finger Repeatability of Robot End-effectors. IEEE Robotics and Automation Letters, 5(2). This author manuscript was made available in PubMed Central (PMC) on November 17, 2020.

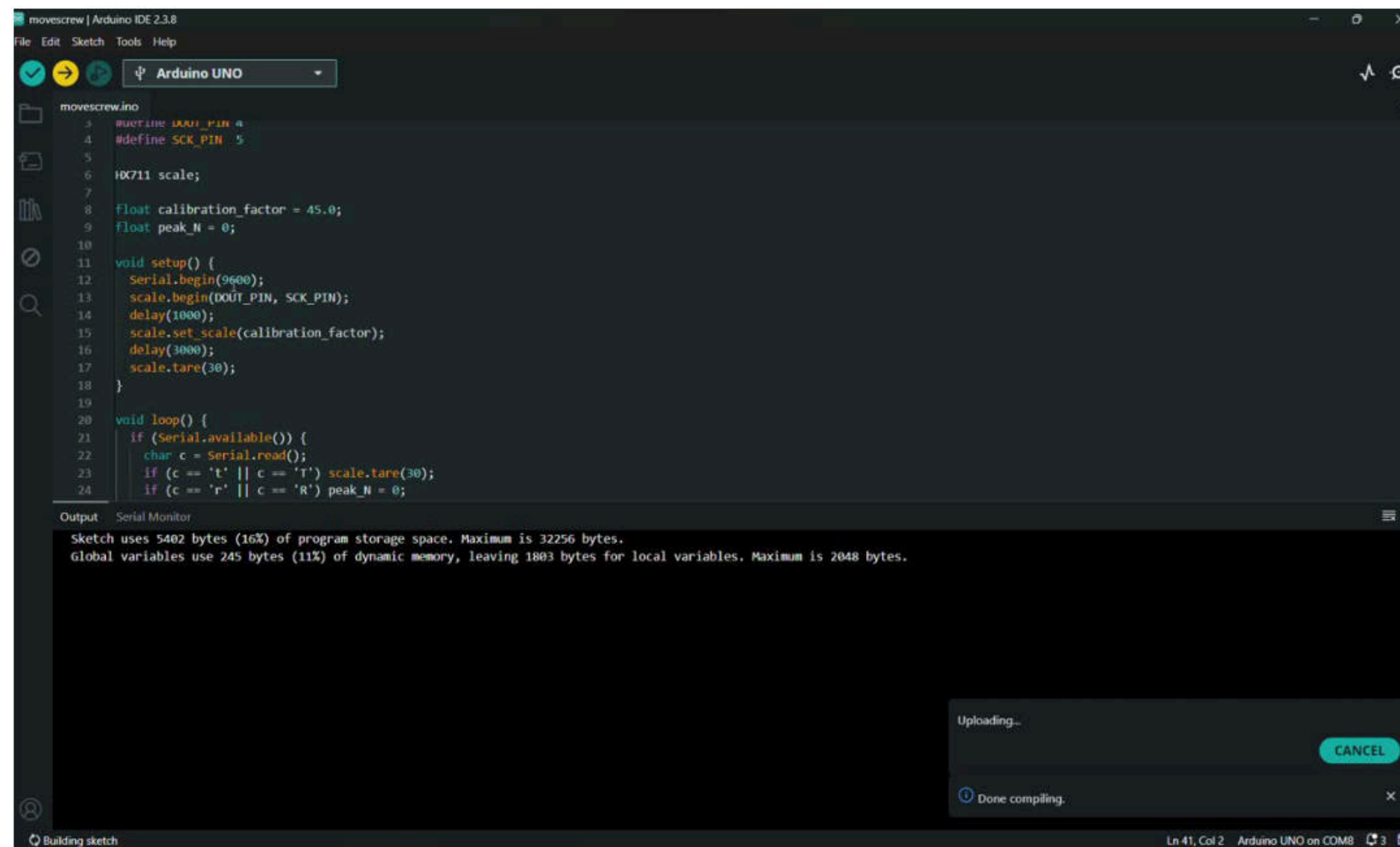
Test Setup



Test Setup



Test Results



```
movescrew.ino
1 #include <Arduino.h>
2 #define DOUT_PIN 4
3 #define SCK_PIN 5
4
5 HX711 scale;
6
7 float calibration_factor = 45.0;
8 float peak_N = 0;
9
10 void setup() {
11   Serial.begin(9600);
12   scale.begin(DOUT_PIN, SCK_PIN);
13   delay(1000);
14   scale.set_scale(calibration_factor);
15   delay(3000);
16   scale.tare(30);
17 }
18
19 void loop() {
20   if (Serial.available()) {
21     char c = Serial.read();
22     if (c == 't' || c == 'T') scale.tare(30);
23     if (c == 'r' || c == 'R') peak_N = 0;
24   }
25 }
```

Output: Serial Monitor

Sketch uses 5402 bytes (16%) of program storage space. Maximum is 32256 bytes.
Global variables use 245 bytes (11%) of dynamic memory, leaving 1803 bytes for local variables. Maximum is 2048 bytes.

Uploading...

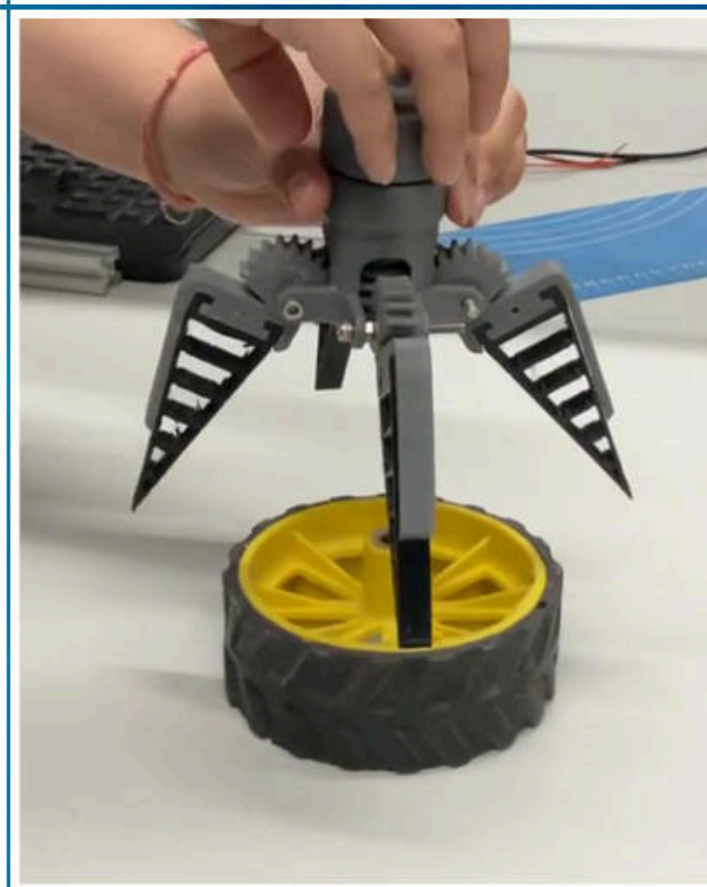
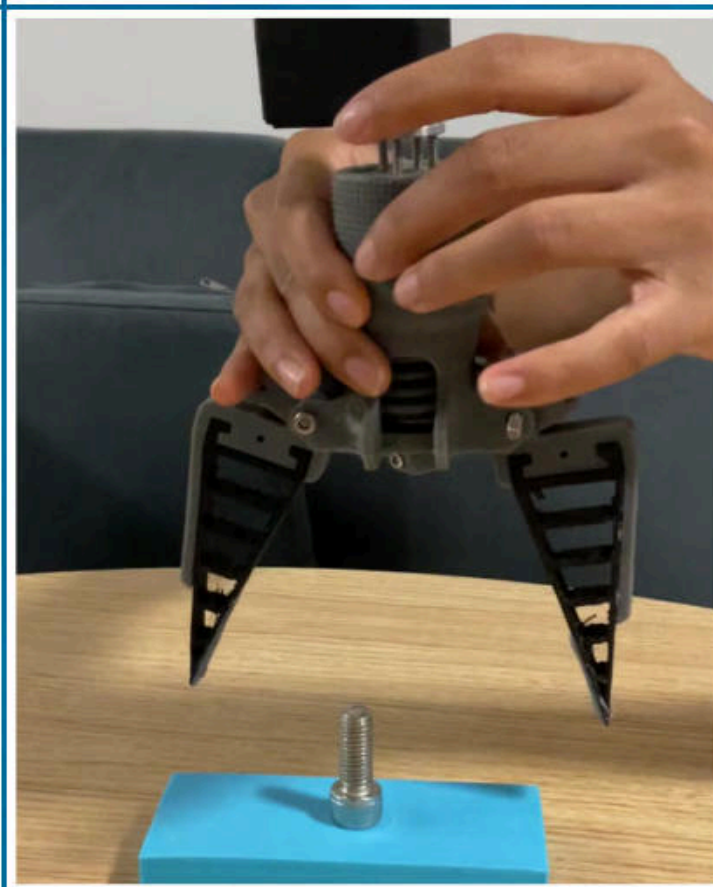
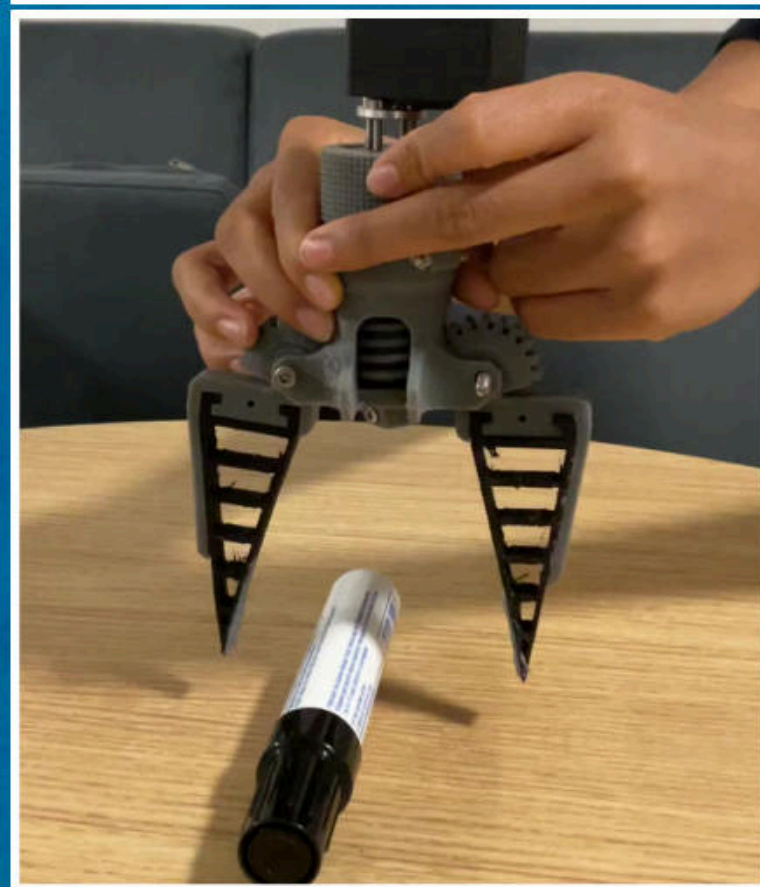
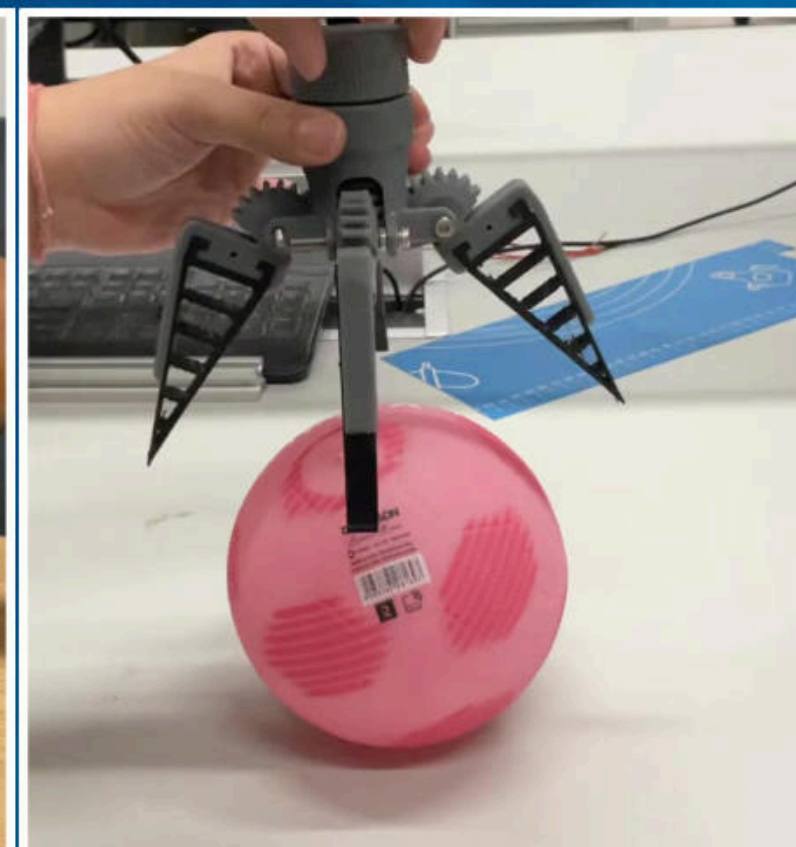
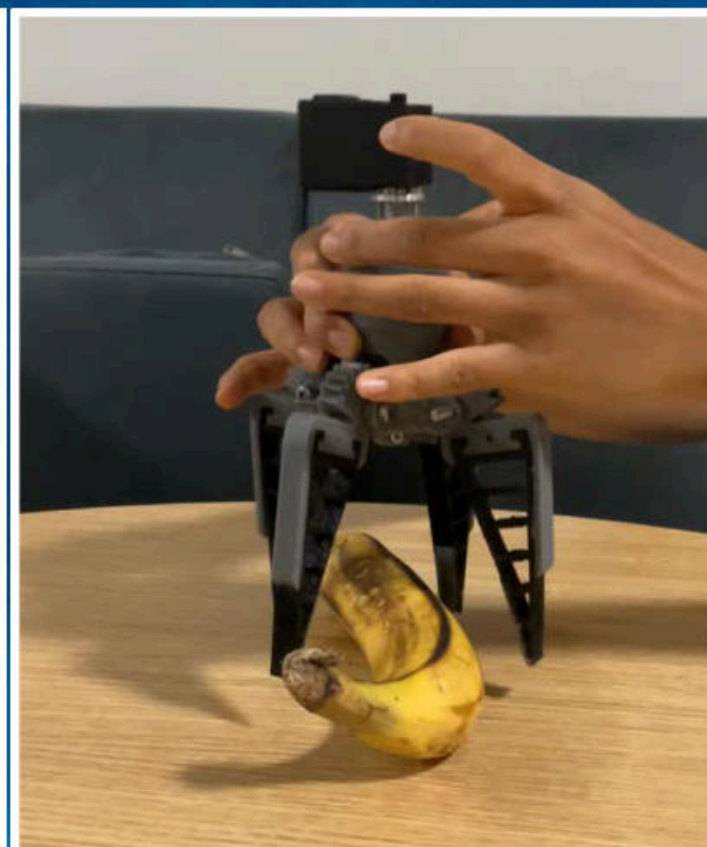
Done compiling.

Result: 3.848 N max strength

Cylinder Diameter (mm)	Hand 1 F_{grasp} (N)		Hand 2 F_{grasp} (N)	
	Mean	95% Confidence Interval	Mean	95% Confidence Interval
50	47.0219	[44.366, 49.468]	118.983	[101.264, 137.843]
80	76.1081	[70.003, 84.323]	92.968	[84.811, 100.669]

Table 2. Mean and 95% confidence intervals of the internal grasp force for Hand 1 and Hand

Testing on Yale-CMU-Berkeley Object set

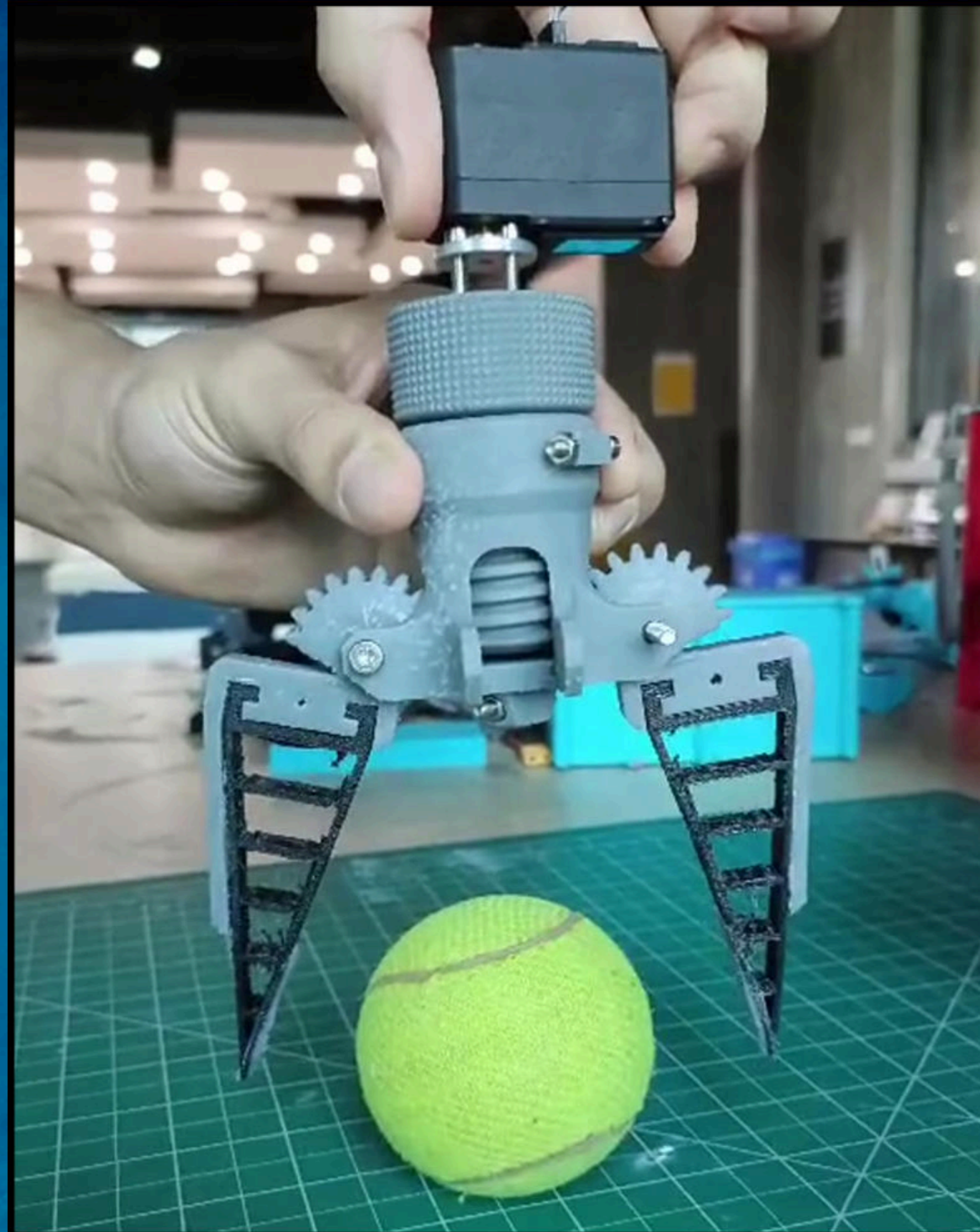


Problem Statement

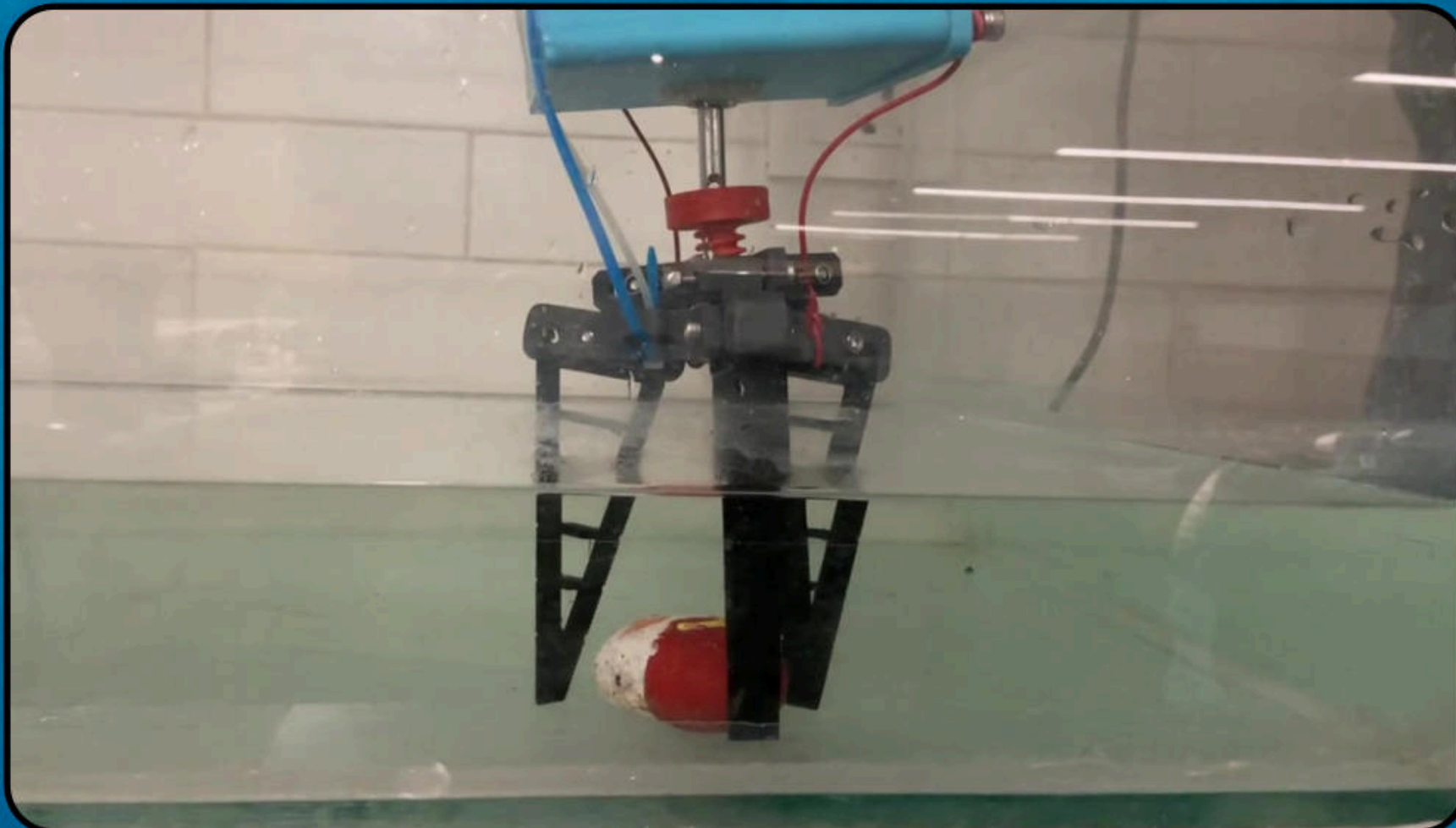
Literature Review

Modelling & Analysis

Testing



Underwater Testing

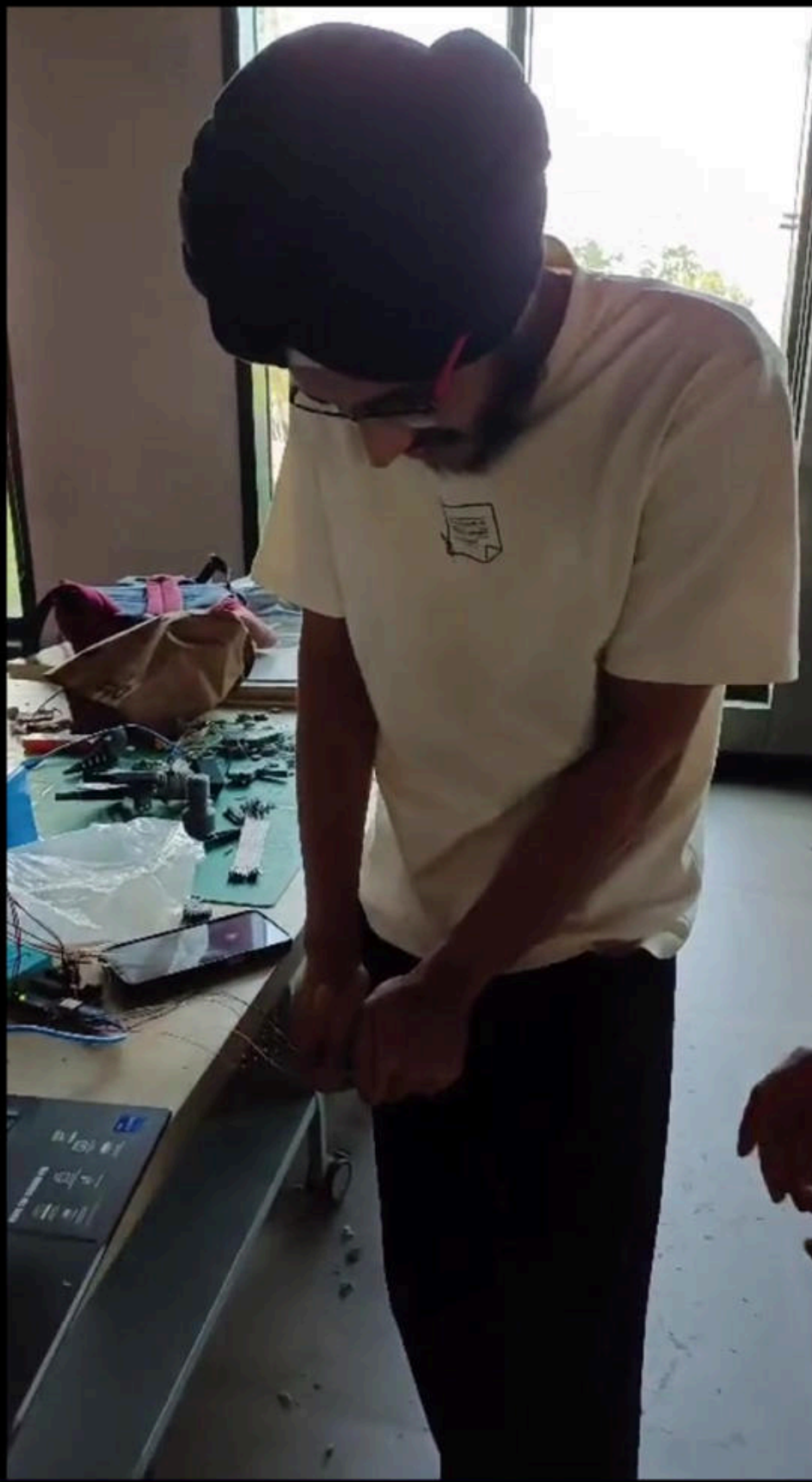


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Thank you.